

THE ICT UNIVERSITY



FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

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Bachelor of Science in Informations System And Networking.

TITLE

**Design and Implementation of a Resilient Backup Network Architecture for
CRTV: Enhancing Operational Continuity and Network Performance**

By

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DECLARATION

The research project entitled “**Design and Implementation of a Resilient Backup Network Architecture for CRTV: Enhancing Operational Continuity and Network Performance**” serves as my independent work and intellectual development. The research work achieves completion as partial fulfillment toward achieving a Bachelor of Science degree in Information Systems and Networking at The ICT University. I confirm this research has not received submission for academic qualifications to any other institution. This work includes proper reference and acknowledgment of all materials along with their sources. Name: NKUMBE LARRY ETAH

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CERTIFICATION

This project titled “**Design and Implementation of a Resilient Backup Network Architecture for CRTV**” has been submitted for review with my consent as the research supervisor.

Engr. **Tekoh Palma**

Signed:

Date:_____

DEDICATION

I dedicate this work to God Almighty for His guidance, wisdom, and strength throughout this journey. Without His grace, this project would not have been possible.

ACKNOWLEDGMENTS

First and foremost, I sincerely thank the Almighty God for His guidance, strength, and grace that enabled me to carry out and complete this project.

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FACULTY APPROVAL

This dissertation has been duly reviewed by the Department and the Faculty and is ready for examination with our approval.

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LIST OF ACRONYMS AND ABBREVIATIONS

Abbreviation	Full Form
CRTV	Cameroon Radio Television
ICT	Information and Communication Technology
DSI	Digital Services and Information
VPN	Virtual Private Network
UPS	Uninterruptible Power Supply
HSRP	Hot Standby Router Protocol
VRRP	Virtual Router Redundancy Protocol
PRTG	Paessler Router Traffic Grapher
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
SDLC	System Development Life Cycle
UML	Unified Modeling Language
GNS3	Graphical Network Simulator-3
SPSS	Statistical Package for the Social Sciences
NAS	Network Attached Storage
SAN	Storage Area Network
AWS	Amazon Web Services
BBC	British Broadcasting Corporation
SABC	South African Broadcasting Corporation

SDN

Software-Defined Networking

IPsec

Internet Protocol Security

ABSTRACT

Cameroon Radio Television (CRTV) relies on a stable and secure network infrastructure to deliver broadcasting and administrative services to the public. However, recurring network failures have led to service disruptions, delayed broadcasts, and communication breakdowns. This project focuses on designing and simulating a resilient backup network system tailored to CRTV's operational environment to maintain uninterrupted services during main network outages. The design process involved analyzing CRTV's existing infrastructure to identify vulnerabilities and selecting appropriate routing protocols, firewalls, and redundant links to enhance network reliability. The proposed backup system was tested in a simulated GNS3 environment to evaluate failover performance, redundancy handling, and routing stability. Results showed successful automatic failover, consistent service delivery during simulated link failures, and improved system resilience. Challenges such as infrastructure limitations, access control, and cyber threats were addressed using firewalls, secure IP schemes, and monitoring tools. Future enhancements include integrating real-time fault detection and smart monitoring features. This backup architecture positions CRTV to deliver continuous, secure, and scalable broadcasting services, even in the face of technical disruptions.

Keywords:

Backup Network, CRTV, Network Resilience, GNS3 Simulation, Redundancy

RÉSUMÉ

La Cameroon Radio Television (CRTV) maintient ses services grâce à une infrastructure réseau qui requiert des conditions de fonctionnement fiables. Cependant, des défaillances fréquentes du réseau entraînent des interruptions de service, des retards dans les diffusions et des problèmes de communication interne. Ce projet vise à concevoir un système de réseau de sauvegarde résilient permettant d'assurer la continuité des opérations de la CRTV en cas de panne du réseau principal. L'équipe de conception a analysé la configuration actuelle du réseau afin d'identifier ses faiblesses et de choisir les outils adéquats pour mettre en œuvre une nouvelle architecture. La solution proposée a été testée dans un environnement de simulation GNS3 pour évaluer ses performances en conditions de panne. La commutation automatique du réseau en cas de défaillance a permis de maintenir la connectivité et la continuité des services. Trois défis majeurs ont été pris en compte : l'accès restreint à certaines infrastructures, les risques de cybersécurité, et la gestion des menaces, qui ont été atténués par des protocoles de sécurité, des pare-feux, et des solutions de sauvegarde des données. Ce système de réseau de secours améliore la fiabilité globale de l'infrastructure, tout en préparant la CRTV à intégrer des outils de surveillance intelligente et de détection automatique des pannes pour un fonctionnement encore plus efficace à l'avenir.

Mots Clés :

Réseau de secours, CRTV, Résilience réseau, GNS3, Redondance

CHAPTER 1: INTRODUCTION

1.1 Introduction

In today's digital world, reliable and uninterrupted network connectivity is critical for the smooth operation of any organization, especially in the media and broadcasting sector. Cameroon Radio Television (CRTV), as the national broadcaster, depends heavily on its IT infrastructure and network systems to deliver timely news, entertainment, and information services to the public. However, the institution has experienced network outages and system failures that disrupt operations and impact service delivery. These challenges highlight the urgent need for a resilient backup network system that ensures continuity even in the face of unexpected failures (Nchinda & Acha, 2021).

A backup network architecture is a secondary infrastructure designed to take over operations in the event that the primary system fails. This type of network plays a vital role in reducing downtime, preserving data integrity, and maintaining business continuity (Kumar & Raj, 2018). Without such a system in place, organizations risk losing valuable information, interrupting communication, and damaging their reputation. For an institution like CRTV, which delivers content to millions of viewers and listeners daily, such disruptions can have significant consequences on public trust and service quality (Mbembe & Tchoumba, 2022).

Network failures may result from various causes, including hardware malfunctions, power outages, cyber-attacks, and software issues (Santos & Lee, 2020). While some of these events are unpredictable, their effects can be minimized with the right backup architecture. A well-planned and well-implemented resilient network will not only prevent total system failure but also improve the overall performance and reliability of the organization's ICT services (Ahmed, Tan, & Zhang, 2020).

Despite the increasing need for resilience, many public institutions, including CRTV, continue to operate with outdated or poorly maintained network infrastructures. In many cases, there is no clear plan for redundancy or recovery in the event of a network failure (Fernandez, Otieno, & Muriuki, 2020). This gap presents a risk to business operations and emphasizes the need for a more robust

solution. Building a backup network system with failover capabilities, secure data paths, and effective recovery mechanisms can provide CRTV with a long-term strategy for operational continuity (Tajudeen, Ngue, & Ako, 2019).

This research aims to design and implement a resilient backup network architecture for CRTV that addresses current challenges, minimizes service interruptions, and enhances network performance. The project involves assessing the existing infrastructure, identifying points of failure, and proposing a new design that includes redundancy, secure protocols, and performance optimization tools. The ultimate goal is to develop a system that can ensure uninterrupted service delivery and improve the reliability of CRTV's operations.

By implementing a resilient backup network, CRTV will be better equipped to handle technical failures and respond quickly to emergencies. This work contributes to the broader goal of digital transformation in public institutions and promotes the use of smart, secure, and efficient technologies in critical sectors (Ola & Adegoke, 2021). Moreover, the findings and lessons from this study can serve as a reference for similar organizations seeking to improve their ICT infrastructure and ensure continuous service delivery.

1.2. Background of the Problem

In today's technology-driven world, reliable network infrastructure is vital for the continuous and efficient operation of any organization. For a public broadcaster like Cameroon Radio Television (CRTV), network availability is not only essential for day-to-day administrative activities but also for the timely delivery of content to millions of viewers and listeners across the country. However, CRTV has experienced several cases of network failures that have led to service disruptions, delayed broadcasts, and data loss, which directly affect its credibility and effectiveness.

Currently, CRTV relies mainly on a single primary network system with limited or no backup mechanisms in place. This setup creates a single point of failure—if any part of the main network goes down due to technical faults, cyber threats, or power interruptions, the entire communication and operational flow is affected. These types of outages reduce productivity, interrupt live and

recorded programs, and may result in the loss of valuable digital content or delays in public service announcements.

The lack of a reliable backup network makes CRTV vulnerable to unplanned downtimes. Most of the network infrastructure was set up without strong redundancy planning, and the existing equipment and configurations are often outdated or not optimized for resilience. In an organization where time-sensitive broadcasting and media services are the core of its operations, such vulnerabilities can be very damaging

Furthermore, CRTV faces challenges related to scalability and performance optimization. As the demand for digital content and multimedia streaming grows, the existing infrastructure struggles to meet the increasing network load. Without an efficient backup system, any effort to scale services or upgrade systems may result in longer downtimes, service interruptions, or system failures.

Security is another major concern. With no isolated backup system, all services depend on the same vulnerable structure. A cyberattack or malware outbreak can easily cripple the whole network, and recovery in such a setup is slow and often incomplete. In today's environment, where cyber threats are becoming more sophisticated, having a secure and resilient network is not optional — it is a necessity.

These issues show the urgent need for CRTV to adopt a modern, resilient backup network architecture. Such a system would provide alternative routes and resources that automatically take over during failures, allowing services to continue without major interruptions. By designing a network that includes failover systems, redundancy in connectivity, secure backup paths, and monitoring tools, CRTV can improve its overall network reliability and performance.

This project aims to address the current weaknesses by proposing and implementing a resilient backup network tailored to CRTV's environment. The goal is to ensure that CRTV remains functional even during system failures and to improve overall service delivery through better network planning and infrastructure management. Solving this problem is key not only to

improving CRTV's operations but also to setting an example for other public institutions facing similar challenges.

1.3. Problem Statement

Cameroon Radio Television (CRTV), as the national broadcasting corporation, relies heavily on its network infrastructure to deliver continuous media content and carry out critical administrative operations. However, the absence of a reliable backup network system has exposed CRTV to frequent service disruptions caused by power outages, equipment failures, cyber threats, and system overloads. These disruptions affect not only internal communication and data flow but also broadcasting services, leading to program delays and loss of public trust (Fernandez, Otieno, & Muriuki, 2020).

The current network setup at CRTV lacks resilience features such as redundancy, automated failover, and load balancing. In the event of a network failure, there are limited alternatives in place to maintain service continuity. This means that even a minor technical issue can bring down important systems, halt media transmissions, and interrupt operations for extended periods. Such downtime can be extremely costly for a national broadcaster whose responsibility is to inform, educate, and entertain the public in real-time (Ahmed, Tan, & Zhang, 2020)..

Moreover, the existing infrastructure is not well-optimized to handle future scalability or modern network demands. As CRTV expands its digital services and media platforms, its network must also evolve to support increased traffic, higher data transmission, and stronger security protocols. Without a well-structured backup system, the institution risks falling behind in service delivery and digital transformation.

Another critical issue is the risk to data integrity and system security. In the absence of a separate backup network or disaster recovery plan, sensitive data and broadcast content are vulnerable to loss or corruption during failures. Additionally, recovering from these incidents is often slow and inefficient due to the lack of pre-configured recovery paths and tools (Kumar & Raj, 2018).

Although resilient network solutions are available on the market, most are designed for private corporations and require expensive technologies or expert-level resources to implement. CRTV, being a public institution with a constrained budget, cannot adopt such high-end systems without customization and cost-effective planning (Tajudeen, Ngue, & Ako, 2019).

This situation presents a major operational challenge for CRTV. The institution's inability to maintain continuous service in the face of network disruptions affects its mission, damages its reputation, and weakens public confidence. Therefore, there is an urgent need to design and implement a resilient backup network architecture tailored to CRTV's needs—one that is scalable, secure, affordable, and efficient.

This study seeks to address these problems by analyzing CRTV's current network environment and developing a backup architecture that can automatically take over during failure, restore services quickly, and enhance overall network performance. By doing so, CRTV will be able to improve service availability, reduce downtime, protect critical data, and strengthen its operational capacity.

1.4. Objectives

1.4.1. General Objective:

This study aims to design and implement a cost-effective and resilient backup network for CRTV to ensure continuous operations during main network failures. CRTV relies heavily on its network for broadcasting and internal communication, but lacks a backup system, leading to service interruptions and data loss. The proposed solution will introduce automated failover features and secure data transmission, focusing on affordability, ease of maintenance, and compatibility with CRTV's current infrastructure. The system will support future scalability, enhanced traffic handling, and incorporate security measures like access controls to protect against misuse and cyber threats, ultimately strengthening CRTV's service reliability and public trust.

1.4.2. Specific Objectives:

1. To analyze CRTV's current network infrastructure and identify key points of failure or vulnerability.
2. To design a resilient backup network architecture that ensures continuous service during main network outages.
3. To implement the proposed backup system using appropriate network technologies and best practices.
4. To evaluate the performance and reliability of the backup network under simulated failure conditions.

1.5. Research Questions

General Research Question

- How can a resilient backup network system improve CRTV's service availability and operational continuity ?

Specific Research Questions

1. What are the key weaknesses and vulnerabilities in CRTV's existing network infrastructure ?
2. What design principles and technologies are most suitable for building a resilient and scalable backup network for CRTV ?
3. How can the implementation of a backup network system at CRTV ensure service continuity during failures ?
4. What challenges might arise during the implementation of the backup network, and what solutions can be applied to address them ?

1.6. Scope of Research

This research focuses on the **design and implementation of a resilient backup network system** specifically for the Cameroon Radio Television (CRTV). The study is centered on addressing the operational challenges caused by network outages, system failures, and the lack of redundancy within CRTV's existing network infrastructure. The main goal is to ensure continuous service delivery, minimize downtime, and enhance the performance and reliability of the organization's network systems.

The project will be limited to CRTV's internal network environment, particularly within the **Digital Services and Information (DSI) department**, which manages most of the institution's ICT infrastructure. The research will assess the current network setup, identify weak points, and propose a backup architecture that includes key elements such as redundancy, automated failover, secure routing, and system monitoring.

This study does **not aim to replace** the entire CRTV network but rather to complement the existing system with a reliable and cost-effective backup solution. The proposed architecture will be designed to function in case of hardware failure, power loss, or other disruptions that may affect the primary network. The solution will consider CRTV's current resources and will be scalable to support future growth and technological advancements.

The platform will be implemented and tested in a **controlled environment** (eg, testbed or simulation lab), as full-scale deployment across CRTV may not be feasible within the timeframe and scope of this project. The study will evaluate the performance of the system under simulated failure conditions to measure response time, failover efficiency, and overall reliability.

This research will not cover the broadcasting equipment or end-user devices such as studio consoles, television sets, or mobile apps. Instead, the focus is on the core network components—routers, switches, access points, servers, and backup links.

Geographically, the study is limited to CRTV headquarters in **Yaoundé**, but the design is intended to be adaptable for use in other regional stations if required in the future. The research does not include external partners or third-party networks connected to CRTV's system.

Finally, the study will not explore broader enterprise-level services such as cloud migration, full disaster recovery frameworks, or cybersecurity auditing. However, basic security features like encrypted communication and access control will be included to ensure the backup network is protected from common threats.

1.7. Significance of the Study

The research proves essential to three main groups as well as technical staff at CRTV and its public service users and public institutions which experience equivalent network problems. CRTV serves the nation through broadcasting information and entertainment together with news to its public audience. The organization's broadcasting delays will increase when network services stop working which reduces the capability to deliver services to the public successfully.

The backup network design process of this study allows CRTV to enhance its service reliability framework. The backup system automatically operates after main network failure because of power cuts or technical malfunctions and cyberattacks to sustain service operations. The backup system functions to lower operational stoppages and safeguard information which ensures CRTV retains public support.

The backup network at CRTV will help technical staff address network problems through better control methods. CRTV can expect benefits of faster recovery and simpler troubleshooting together with improved plans for future upgrades through this network solution. The backup network decreases the need for human manual responses to failures which allows operations to maintain normal service delivery throughout emergencies.

The project provides beneficial support for governmental organization departments as well as policy-making officials. The study demonstrates a network management system applicable to public institutions which depend on reliable network infrastructure such as hospitals ministries and educational institutions. Operating backup systems protects essential services from interruptions and maintains continuous communication throughout emergencies.

The findings from this research promote digital transformation objectives throughout Cameroon. Stable networks of high quality have become essential for organizations that embrace digital

service implementation. The establishment of robust network infrastructure progresses communities toward smart interlinked public institutions.

Finally, the research will contribute to the **academic and technical community** by providing a real-world case by providing a real-world case study of how to design and implement a cost-effective backup network in a resource-constrained environment. It will offer ideas and lessons that can help other students, researchers, and IT professionals improve network reliability in similar situations.

1.8. Limitations of the Study

The research seeks to develop an operational emergency network system for CRTV which enhances network reliability yet several restrictions remain unaddressed.

The study faces a major challenge because CRTV restricts researchers from examining their complete network infrastructure. Research access is restricted exclusively to security policies of the Digital Services and Information (DSI) department as well as organizational constraints. The final solution might show incomplete adaptation to full technical scenarios beyond the DSI department and other CRTV operational departments.

The backup system must undergo testing within a simulated condition instead of the actual CRTV production network environment. During testing the observed performance levels could differ from live operational conditions under real-world demanding environments. A comprehensive deployment needs additional time, required permissions and additional resources than this academic research project can provide.

Financing and research-related resources available to this study are constrained. Student project constraints prevent the study from utilizing professional-grade networking infrastructure or enterprise-level systems which CRTV operates in their day-to-day production. The design implements affordable solutions while using available tools and technologies despite not offering all possible high-end options that can be purchased on the market.

The project needs to fulfill all academic timelines set by the schedule. The testing along with data acquisition and iterative processes suffer restrictions because of these time constraints. Different advanced features which include artificial intelligence-based monitoring as well as full redundancy

automation and multi-location synchronization will be suggested in this phase though their thorough development and testing remains incomplete.

This research deals exclusively with understanding network infrastructure that includes backup routing elements consisting of routers switches links and communication protocols. Other essential systems managed by the corporation remain outside the examination including broadcasting equipment, audio/video management, studio operations and mobile content delivery platforms. The design phase does not include analysis of system components outside network infrastructure.

Finally, this research is **limited to CRTV's headquarters in Yaoundé** . While the architecture is designed to be scalable and reusable, its adaptability to CRTV's regional centers or to other public institutions in Cameroon or Africa would need separate testing and validation.

Despite these limitations, the study provides a strong foundation for future improvements and sets a practical direction for CRTV to follow in strengthening its ICT infrastructure.

1.9. Organization of the Study

The research project is structured into five chapters that collectively explore the design and implementation of a resilient backup network for CRTV. Chapter One introduces the study by outlining the background, problem statement, research objectives, and questions, along with the scope, significance, and limitations of the research, providing a foundational understanding of the need for a backup network at CRTV. Chapter Two presents a literature review that evaluates existing studies on network reliability, redundancy systems, and backup strategies within public institutions, identifying knowledge gaps and how this study aims to address them. Chapter Three outlines the research methodology, detailing the data collection techniques, analysis methods, tools, and technologies used to assess CRTV's current network and develop the proposed solution. Chapter Four focuses on the implementation of the backup network, including the system architecture, design diagrams, selected tools, and simulation and testing procedures used to evaluate the network's performance. Finally, Chapter Five summarizes the key findings, draws conclusions

based on the research results, and offers recommendations for CRTV and similar institutions, while also suggesting future areas of study.

CHAPTER 2: LITERATURE REVIEW

2.1. Introduction

In any research undertaking, an extensive literature review is an essential requirement to the success of the undertaking. This part reviews prior research on network resilience and backup systems uncovering challenges and solutions that guarantee continuity of CRTV service despite system failures (Ahmed, Tan, & Zhang, 2020).

CRTV is the national broadcaster of Cameroon who highly depends on the network to facilitate the internal communications, broadcasting public announcements, and the delivery of media contents. However, the absence of a trusted backup system often leads to service interruption during broadcast lags, data loss, and poor performance (Nchinda & Acha.

This part focuses on the critical demand for a robust network infrastructure on the side of public service organizations. Outages are normally as a result of power outages, device error, network resilience achieved by proactive identification of these threats and quick recovery mechanisms in place.

The review examines the most important technologies used in backup networks such as failover systems, redundant links, load balancing, and automatic switching strategies and their capability of supporting continuous service. These solutions

In developing countries, the efforts to improve networks are often restricted by a lack of budget, aging infrastructure, and lack of knowledgencimals which need to be managed especially upon introducing backup networks with the aim of protecting systems from the invasion and attacks (Ola & Adeg

Although there is abundant global research on network design and data recovery, there are relative few studies addressing the field in the case of public broadcasting systems in Africa.

For the purposes of this review research draws on various material including scholarly articles and technical documents, as well as case studies and field implementation from similar organizations.

Preference is given to solutions that not only provide economic efficiency but also have a scalable solution ideal for CRTV's unique operations.

This chapter finally highlights the role of backup networks, identifies corresponding technologies, and reviews world best practices, taking Cameroon's unique environment into consideration. These findings can guide the creation of a strong backup network at CRTV with a view to enhancing service reliability and raising public confidence.

2.2. Overview of Network Architecture in Public Institutions

General operations of national broadcasters like Cameroon Radio Television (CRTV) depend heavily on network architecture to maintain their smooth running. The structure of a network creates conditions where departments can exchange data and gain application access together with efficient communication systems to provide uninterrupted public services. The role of network stability at CRTV becomes crucial because this broadcaster needs to deliver news broadcasts and national programs. Network failures trigger multiple problems that include broadcast delays, disrupted internal communication, and stopped content distribution (Nchinda &

The network architecture of public institutions consists of routers, switches, servers, firewalls, and access points, which enable seamless operations between internal staff and external actors. The implementation of cloud services together with secure VPNs in contemporary setups allows institutions to boost accessibility while enhancing flexibility (Ahmed, Tan, & Zhang, 2020). The majority of public organizations face infrastructure problems while struggling to implement sufficient backup solutions due to outdated systems (Mbembe & Tchoumba, 2022). These institutions remain exposed to disruptions caused by power outages, equipment malfunctions, and online security threats, which significantly endanger operational continuity (Kumar &

CRTV experiences similar development challenges as other public institutions in developing countries, marked by limited funding, inadequate backup systems, and technology obsolescence. CRTV's information network relies on a single primary pathway; hence, a breakdown of any critical

device or failed link can lead to total system downtime. This **single point of failure** poses a serious risk and exposes the

A resilient network framework is necessary to address these

In addition to technical setbacks, public institutions face administrative challenges

Security remains one of the most pressing concerns. Public networks handle sensitive data—employee records, financial details, and confidential government communications—which, if compromised, can result in serious breaches. Without proper backup and recovery systems, institutions risk catastrophic data loss due to system failures or cyberattacks (Ola & Adegoke, 2021). A secure and resilient network must therefore integrate firewalls, access control,

Across Africa and other parts of the world, some public institutions have begun deploying resilient network systems, yielding improved service uptime, enhanced security, and rapid disaster recovery (Fernandez, Otieno, & Muriuki,

The need for uninterrupted public service at CRTV makes deploying a resilient network more than a technical upgrade—it is a critical operational necessity. This project will design a backup network system tailored specifically to CRTV's needs and constraints, enabling the organization to improve reliability, reduce downtime, and deliver more consistent services to the public.

2.3. Understanding Network Failures and Downtime

Network failure is one of the most common and serious problems faced by organizations that rely on digital infrastructure for their daily operations. For public institutions like **Cameroon Radio Television (CRTV)**, which depends heavily on a stable network to broadcast programs, manage data, and communicate internally, any form of network failure can lead to major disruptions. These outages not only affect service delivery but also damage the organization's public image and reduce operational efficiency.

Network downtime refers to periods when the system or services are unavailable due to failures in the infrastructure. These failures may be caused by **hardware breakdowns, power outages, software errors, human mistakes, cyberattacks, or overloaded systems**. CRTV has experienced

these issues in the past, leading to delayed broadcasts, communication breakdowns, and in some cases, complete halts in operations.

In many public institutions, including CRTV, the main problem lies in the **lack of redundancy** and **poor planning for backup systems**. Most of the network infrastructure is designed as a single path system. This means if one critical component—like a router or switch—fails, the entire system can crash. Without a secondary network to take over, the recovery process becomes slow, manual, and often complicated.

Another common cause of failure is **poor maintenance** of network equipment. Over time, outdated routers, switches, and cables may become unstable or incompatible with new systems. CRTV, like many government agencies, operates within a tight budget and may delay upgrades, which increases the risk of breakdowns.

Power outages are also a major issue in Cameroon. Even with Uninterruptible Power Supplies (UPS) or generators, sudden shutdowns can still affect networking equipment. If backup power is not properly linked to the network infrastructure, services can go offline instantly during a blackout.

Cybersecurity threats also play a big role in network failures. Viruses, ransomware, and Distributed Denial of Service (DDoS) attacks can overwhelm or shut down systems. Without proper security protocols, the entire network can be compromised, and it may take hours or days to restore.

One of the most damaging aspects of network failure is the **loss of data** or inability to access important files. In broadcasting, this may include audio or video files, live feeds, or scripts that are scheduled to air. Without a reliable backup system, CRTV risks losing critical content or being forced to postpone programs, which affects public trust and viewership.

To avoid these issues, organizations need to design **resilient network systems** that include automatic failover, redundant paths, and regular data backups. A resilient network does not stop working when one part fails. Instead, it shifts the traffic or services to a secondary system that keeps operations running. This is especially important for institutions like CRTV that provide 24/7 services to the public.

In developed countries, most large organizations already use **network resilience features** like load balancing, link aggregation, dual routers, and cloud-based backups. However, many public institutions in developing countries still rely on outdated or single-point systems that lack these capabilities. This makes them more vulnerable to failure.

This section has shown that understanding the causes of network failure is the first step toward building a better system. CRTV needs a well-planned **backup network architecture** that can support its operations even when the main system fails. In the next sections of this chapter, the focus will shift to the technologies and methods that can be used to design and implement such a system, along with examples of how other institutions have solved similar problems.

2.4. Concepts of Network Redundancy and Resilience

The construction of reliable and stable IT infrastructure depends heavily on the concepts of redundancy and resilience for network management. The implementation of these concepts enables organizations to prevent long disruptions in service in case systems experience failures from power outages or hardware malfunctions or cyberattacks. The essential nature of a resilient and redundant network system exists for Cameroon Radio Television (CRTV) because the organization depends heavily on broadcasting continuity and data accessibility.

A redundant network contains additional paths or devices together with systems which activate themselves to take over from the primary system when it stops working. If network failures occur backup routers as well as switches and links function immediately to restore functionality. Redundant networks include fail-safe architecture which enables automatic service continuity when parts of the system fail since backup systems immediately take over.

Network resilience describes how well a system functions alongside its quick recovery from failure. The essential characteristics of a resilient network consist of automatic problem detection with simultaneous traffic redirection through alternate paths to restore normal operations without human intervention. Systems with network resilience possess hardware resilience through increased device and link provision and power supply resources alongside software resilience through automated monitoring systems and failover protocols alongside alert functions.

Most network operations at CRTV function through a solitary transmission path. One failure of a critical router or switch possesses the ability to disrupt most of CRTV's services. The present configuration presents significant hazards to all institutions particularly organizations which distribute real-time information because it functions through a single path. CRTV remains susceptible to operational interruptions along with damaged reputation because it lacks proper back-up systems and redundancy capabilities.

The solution requires backup networks to have multiple redundant components. These can include:

1. Internet service backup connections from different service providers as a protection against network failures.
2. A network operates with automatic failover capabilities through protocols HSRP or VRRP which handle redundant routing paths.
3. Whenever power outages occur backup power systems consisting of generators and UPS will operate critical devices.
4. Network devices such as routers and switches need absolute readiness to perform as backup systems.

Network resilience depends on installation of proper monitoring tools which identify failures before executing automatic backups. Network monitoring tools PRTG, SolarWinds and open-source alternatives enable administrators to monitor network traffic while producing raw data for detecting bottlenecks before complete failure occurs.

Resilience contains security protections as one of its essential components. Put in place measures to defend backup systems since they require identical protection as main networks. The combination of firewalls together with access control measures and data encryption defenses provides protection to running and backup network traffic.

Large organizations along with critical infrastructure facilities view network resilience as a fundamental necessity for operation. Multiple media outlets along with public broadcasters in developed countries maintain backup networks which provide nonstop service operations. The

widespread adoption of redundant communication frameworks in Cameroon remains scarce because of limited organizational budgets and insufficient personnel skills as well as insufficient planning efforts.

Organization leaders are beginning to establish resilience through secondary network implementations as well as off-site backup plans with failover systems. Limited resources cannot prevent reliable network creation because correctly designed scalable and managed systems succeed in such conditions.

The implementation of resilient backup networks by CRTV will enhance service reliability because it reduces downtime and ensures important data protection. The broadcasting operations can remain active when part of the system fails through this setup since it protects public service obligations of the institution.

This section highlights why redundancy and resilience are critical in network planning. The next part of the literature review will explore the specific tools and technologies used in building such systems, including both hardware and software components suitable for CRTV's environment.

2.5. Technologies Used in Backup Network Design

Building a resilient backup network requires using the right technologies to ensure that services continue running even when part of the system fails. For an organization like **Cameroon Radio Television (CRTV)**, which operates 24/7 and depends on constant network availability, these technologies are key to reducing downtime and improving service delivery.

One of the most important technologies in backup network design is the use of **redundant devices and connections**. This means having backup routers, switches, and network paths that automatically take over when the primary ones fail. Protocols like **Hot Standby Router Protocol (HSRP)** and **Virtual Router Redundancy Protocol (VRRP)** are used to manage this automatic switching. These tools make sure that users don't notice any service interruption when a failure occurs.

Load balancers are also important in backup networks. They distribute network traffic evenly across multiple servers or links. If one server becomes overloaded or fails, the load balancer

redirects the traffic to another healthy server. This helps keep the system stable and avoids sudden crashes caused by too much traffic on one device.

Network monitoring tools such as **PRTG**, **Nagios**, or **SolarWinds** are used to detect problems early. These tools send alerts when something goes wrong, allowing the IT team to act quickly. In more advanced systems, the monitoring tools can even trigger automated recovery processes without human intervention.

Another useful technology is **link aggregation**, also known as bonding or teaming. This allows multiple network cables or connections to work together as one. If one link goes down, the other links keep working, so data still flows without interruption. This is especially useful in high-bandwidth environments like broadcasting centers.

Cloud-based storage and backup services can also be part of a resilient network system. These platforms store important data and system backups off-site. In case of a disaster or local failure, CTRV can restore lost data from the cloud. Services like **Google Cloud**, **Microsoft Azure**, or **Amazon AWS** offer secure and scalable backup options.

For internal recovery and data protection, **Network Attached Storage (NAS)** or **Storage Area Networks (SAN)** are often used. These systems are designed to store large amounts of data and provide quick access during recovery. They also support regular automatic backups to prevent data loss.

To protect against **cyberattacks**, **firewalls**, **intrusion detection systems (IDS)**, and **virtual private networks (VPNs)** are used. These security tools ensure that both the main and backup networks are protected from unauthorized access and malware. In a broadcasting organization, keeping data and live feeds secure is critical.

Power backup technologies are also essential. **Uninterruptible Power Supplies (UPS)** and **standby generators** are used to keep network devices running during power outages. If CTRV's main power source fails, these systems provide enough electricity to keep the network online or allow it to shut down safely.

Another modern approach is the use of **Software-Defined Networking (SDN)**. SDN allows network administrators to manage network traffic using software instead of physical hardware. This makes the network more flexible and easier to reconfigure in case of changes or failures.

Lastly, **virtualization tools** like **VMware** or **Hyper-V** allow CRTV to run multiple systems on a single server. This reduces hardware costs and makes it easier to switch from one system to another during failure.

In summary, building a reliable backup network system for CRTV involves combining several technologies—redundant devices, load balancing, automated monitoring, power backups, cloud services, and security systems. When used together, these technologies help create a network that is stable, secure, and ready to respond during failure. In the next section, the study will look at real-world examples where these technologies have been used successfully in similar institutions.

2.6. Case Studies on Resilient Networks in Media and Government Sectors

Media and government institutions around the world have faced serious challenges due to network failures, which affect their operations, service delivery, and public communication. To solve these problems, many of them have invested in **resilient backup network systems** that help reduce downtime and improve reliability. Studying these real-life examples can help **CRTV** understand how to build a system that fits its needs and environment.

One good example is the **British Broadcasting Corporation (BBC)** in the United Kingdom. The BBC uses a multi-layer network structure with full redundancy at all major sites. If one data center or broadcast link fails, another takes over automatically. Their system includes load balancers, duplicate routers, multiple internet service providers, and continuous monitoring tools. This setup helps them stay live even during hardware or power failures.

In **South Africa**, the **South African Broadcasting Corporation (SABC)** upgraded its infrastructure to include fiber optic links and redundant core switches to improve stability. They introduced virtualized servers and remote monitoring, which allow them to quickly detect problems and switch to backup systems without affecting broadcasts. Their system was built in phases, making it affordable while still improving resilience over time.

Another useful case is the **Government of Estonia**, known for its advanced digital systems. Estonia's government institutions are connected through a national secure data exchange network called **X-Road**. This system has backup data centers in different locations, constant data replication, and encrypted traffic. Even if one server is attacked or fails, the services continue without interruption.

In **Kenya**, some government offices have implemented hybrid cloud solutions. By combining on-premise servers with cloud-based backup, they ensure that critical data and services can be recovered easily. These systems also include power backups and real-time monitoring, which make recovery faster during outages.

While these institutions are in different countries and may have more resources, the key takeaway is that **resilient networks are built step by step using practical, available technologies**. What matters most is the design—using redundancy, monitoring, power backup, and automatic failover to avoid full system failure.

For **CRTV**, these case studies show that building a reliable backup network is possible, even in environments with limited resources. By learning from others and using scalable tools, CRTV can improve its network and reduce the risk of service interruption. The next section will look at common challenges that might come up during the implementation process and how to address them effectively.

2.6.1. Lessons Learned and Relevance to CRTV

The previous section's case studies show important lessons which CRTV can use when they aim to enhance their network infrastructure. Various media and governmental institutions have developed practices which enable downtime reductions while sharpening service delivery and establishing network resilience for building trust.

Network redundancy represents one of the main lessons organizations should learn. The BBC and SABC among other organizations have developed dual network pathways and automatic system transition systems which enable continued operations during main network failures. CRTV needs to avoid depending on one internet route or service provider in its network operations. CRTV needs

to establish several network connections through failover protocols including HSRP and VRRP because these configurations enable continuous operation when network disruptions occur.

The essential lesson learned by CRTV must include monitoring operations in real-time through fast adoption of response protocols. Strongly resilient institutions maintain alert systems which notify their network administrators about any detected faults. Fast action becomes easier after downtime reduction from this approach. Through implementation of monitoring tools PRTG or SolarWinds CRTV will be able to detect early system failures while their backup systems activate automatically.

The main learning points involve data protection together with security measures. The secure communication network together with encrypted backup systems in Estonia protects services and data from system failure and cyberattacks. CRTV should implement better firewalls and secure remote access capabilities and establish planned scheduled backups to defend against sensitive content threats and internal communications breaches.

The planned implementation approach was adopted by multiple case studies. Multiple institutions began their transformation by selecting one department and extending their efforts across multiple departments before general implementation. CRTV should start by deploying the backup network within its DSI (Digital Services and Information) division before extending it throughout all organizational departments.

The most important lesson of resilient networks shows that costly solutions are not essential for attaining network resilience. Successful network operations depend on planning with smart approaches and choosing correct tools while maintaining the network at optimal condition. CRTV should select open-source software and affordable devices together with cloud services which deliver outstanding performance at cost-effective rates.

These lessons demonstrate that CRTV can enhance its network reliability when investing into redundant systems as well as monitoring programs and enhanced security measures and deployment methods following systematic principles. Through learning from others CRTV can develop an improved system which supports their public service elements by avoiding typical operational missteps.

2.6.2. Operational Challenges Faced by CRTV

Just like other public institutions in developing countries, **CRTV** faces several operational challenges that affect the smooth running of its network systems. One of the biggest issues is the **lack of modern technical infrastructure**. Much of the network equipment in use is outdated and not built to handle today's demands such as live streaming, high data traffic, or backup switching. This makes the system slow and vulnerable to failures, especially when no backup paths are available.

Cybersecurity is also a major concern. CRTV handles sensitive internal data and live content. Without strong protection like firewalls, intrusion detection, or regular updates, the network becomes an easy target for attacks. Many public institutions lack the trained staff or budget needed to manage these risks effectively, which puts the entire system at risk.

Another big challenge is **limited funding**. Designing and setting up a resilient backup network requires investment in hardware, software, training, and ongoing maintenance. For CRTV, with a fixed public budget, finding funds for large upgrades can be difficult.

There's also the issue of **technical skills**. Some staff members may not have the training needed to install or manage backup systems. This can lead to errors, delays, and even system misuse. Without regular training, even a good backup design may not work as expected.

Lastly, **energy supply issues**, like frequent power cuts, also affect network performance. If backup power systems like UPS or generators are missing or not maintained, the network can go down completely during an outage.

These challenges make it clear that CRTV needs a carefully planned solution that fits its technical and financial environment. A simple, secure, and scalable backup network can help address these problems and improve service continuity.

2.6.3. Impact of Operational Constraints on Backup Network Adoption at CRTV

The development of CRTV's resilient backup network faces problems from both governmental regulations and operational limitations. CRTV needs to purchase compliance systems before

building new infrastructure because government standards and data processing rules are tightly enforced. The existence of regulation forces CRTV to redirect essential funding which should be utilized for critical network improvements.

The unreliability of both internet and power supplies in Yaoundé hinders the deployment of cloud backup systems and online system monitoring initiatives. Staff members do not have sufficient training to operate automated failover systems while engineering staff demonstrates little knowledge of new networking tools.

Various elements that cause command and control system improvements to fail together create doubt about new backup systems. Proper maintenance and testing of the failover system is crucial because inadequate testing can trigger service interruption during actual outages thus damaging CRTV's public image. Information loss that occurs with a switch-over operation will cause viewers to doubt the broadcaster's reliability as it pertains to maintaining stable service.

To overcome these barriers, CRTV can:

- **Partner with local ISPs or tech firms** for better connectivity and technical support.
- **Invest in targeted staff training**, focusing on network monitoring and failover procedures.
- **Phase in backup components**, starting with critical links and power backups before full network-wide rollout.

By addressing both the regulatory requirements and on-the-ground challenges, CRTV can successfully implement a backup network that boosts uptime and public trust.

2.7 Case Studies on Resilient Networks

Resilient network systems have helped many public and media organizations maintain stable operations and avoid downtime. In this section, we look at real-life examples of how other institutions have successfully designed and used backup network solutions. These case studies help us understand what worked, what challenges were faced, and how similar methods can be applied to **CRTV**.

By studying these cases, CRTV can learn **best practices** such as using automatic failover systems, redundant network paths, cloud backups, and secure monitoring tools. These examples provide useful guidance for building a strong and affordable backup network that fits CRTV's structure and technical environment.

Case Study 1: **BBC's Network Resilience**

Overview

The British Broadcasting Corporation (BBC) operates a multipurpose network system featuring complete backup capabilities. The BBC operates parallel data facilities between London and Salford which duplicate their routers and switches and internet connectivity between the sites. The system relies on HSRP protocols for failover automation and load balancer services that perform traffic distribution.

Impact

Live broadcasting of BBC programs remains operational when any single location in the network faces a system failure. The automated fault detection system eliminates the need for human operators because it initiates traffic redirection which maintains broadcast quality and causes downtime less than one minute.

Relevance to CRTV

The main headquarters of CRTV should integrate with a secondary data hub through a multi-site solution. Automatic failover protocols with load balancing will ensure CRTV broadcasting stays operational when facing local outages. CRTV can begin its redundancy efforts by creating duplicate routing links for essential sites which it will expand across its networks progressively.

2.8 . Security Considerations in Backup Network Design

The review confirms that protecting backup networks holds the same significance as implementing redundancy. The essential security protocols for CRTV feature the following aspects:

1. Embed IPsec tunnels or use VPNs between primary and backup sites in order to defend data transmission during transit.
2. Strong Access Controls: Implement role-based permissions and two-factor authentication for all network devices.
3. System administrators must practice regular firmware and software updates across all their routers switches and firewalls because this practice eliminates known security threats.
4. Security systems equipped with IDS/IPS sensors should monitor live and backup network paths for detecting and blocking security incidents prior to their spread.
5. Security-Centric Backup Storage Consists of encrypting backup configurations and vital data which should exist off-site or within hardened cloud infrastructure.
6. The collection of network element logs should be centralized into one place for regular examinations that seek out suspicious system activity.
7. The organization needs to perform Disaster-Recovery Drills during scheduled tests which simulate realistic conditions to validate the security policies which function during network failovers.

The backup network design of CRTV that combines these security practices will protect both functional outages and cyber threats to its secondary systems..

2.8.1. Summary of Key Insights

1. Importance of Network Resilience

Public broadcasters like CRTV need continuous network access to deliver live content and manage operations. Even short outages harm service delivery and public trust.

2. Redundancy and Failover

Backup links, duplicate routers/switches, and automatic failover protocols (e.g., HSRP, VRRP) are essential to ensure there is no single point of failure.

3. **Critical Technologies**

Load balancers, link aggregation, cloud-based backups, and network monitoring tools help distribute traffic, secure data, and detect failures early.

4. **Security Measures**

Encrypted tunnels, strong access controls, regular patching, and IDS/IPS systems protect both primary and backup networks from cyber threats.

5. **Implementation Challenges**

Limited budgets, outdated equipment, unreliable power/internet, and low technical skills can slow down deployment and maintenance of resilient networks.

6. **Best Practices from Case Studies**

Phased rollouts, multi-site architectures, off-site backups, and partnerships with local ISPs or tech providers offer practical, cost-effective ways to build and manage a backup network

2.8.2. Identified Research Gaps

The current network resilience and backup system literature shows several deficiencies especially when applied to CRTV's broadcasting platform in Cameroon.

1) Limited Local Studies

Research investigation mostly concentrates on developed nation institutions and large private corporations. The research on backup network designs for public broadcasters operating in settings with limited resources remains sparse particularly for an organization like CRTV.

2) Adaptation of Advanced Protocols

The documentation of protocols like HSRP and VRRP along with SDN-based failover exists extensively but research about their affordable implementation in budget-constrained environments without stable power remains scarce.

3) Practical Deployment Strategies

Multiple theories about redundancy and resilience exist but practical step-by-step frameworks for deployment remain scarce because they need to account for CRTV's existing infrastructure and staff skills together with their approval procedures.

4) Integrated Security in Failover Systems

The subject of network security exists independently from backup resilience despite numerous research studies in both areas. A unified system design must integrate failover defenses with authentication security while securing backup protocols.

5) User-Centered Maintenance Models

Backup networks need regular checks alongside dedicated support procedures for their proper operation. Public institutions lack sufficient research about teaching methods to effectively maintain and resolve problems in backup systems with their non-specialist personnel.

2.8.3. How This Study Addresses the Gaps

This research fills the identified gaps by developing a backup network architecture specifically for **CRTV**. Key contributions include:

1. **Local Adaptation**

The design is based on CRTV's existing infrastructure, power conditions, and budget constraints in Cameroon—ensuring practical deployment and maintenance.

2. **Affordable Resilience Technologies**

We integrate cost-effective protocols (HSRP/VRRP), link aggregation, and open-source monitoring tools to achieve redundancy and automatic failover without high licensing fees.

3. **Operational Roadmap**

A step-by-step implementation guide addresses approval processes, phased deployment in the DSI department, and staff training—making the transition smoother and faster.

4. **Built-in Security**

The architecture combines encrypted VPN tunnels, role-based access controls, and regular patch schedules so that both primary and backup networks stay protected during failover.

5. Scalability and Maintainability

Modular designs allow CRTV to start with critical links and expand redundancy over time. Simple maintenance procedures empower existing IT staff to manage the system confidently.

6. Alignment with CRTV's Mandate

By ensuring uninterrupted broadcasts and data services, the solution directly supports CRTV's public service mission and strengthens viewer trust in national media.

This review highlights how resilient network architectures can keep critical services online during failures, yet shows a lack of practical, low-cost designs for public broadcasters in settings like CRTV. Key gaps include adapting proven protocols to local power and budget constraints, integrating security into failover systems, and providing clear implementation steps for in-house teams.

By drawing on global best practices and tailoring them to CRTV's environment, this study will propose a backup network design that is affordable, secure, and easy to maintain. In doing so, it lays the foundation for CRTV to improve its uptime, protect its data, and continue delivering reliable broadcasts to the public.

CHAPTER 3: METHODOLOGY

3.1 Introduction

The following section of the research outlines the approaches used to design an accessible and reliable backup network appropriate for Cameroon Radio Television (CRTV). The aim is to integrate an efficient and low cost backup network that is tailored to CRTV's operation needs. This chapter describes the way one collects relevant information, creates the network blueprint, and implements the proposed resolution, insisting on both technical good sense and practical usability (Kumar & Raj).

This research uses an implementation-oriented (research) approach aimed at meeting CRTV's performance and continuity requirements. In particular, three basic questions are the focus of the research:

1. What methods of gathering information were used about the problems of CRTV's network backup system?
2. Process of data analysis helped gaining insight into current problems, which was further used to develop precise and pragmatic backup strategy for CRTV.
3. What design and simulation methods were used to develop the backup network plan?

Interviews, document studies, and on-site visits in CRTV's technical department constituted the basis for data collection. These research methods allowed researchers to explain the network's configuration, establish the main sources of system outages, and appraise the available resources and constraints (N

To collect input the study made use of a convenient sample and contacted CRTV's IT personnel that is technicians and network engineers. As they directly manage and maintain the network, the IT staff gave critical insight into operation problems and needs (Mbembe & Tchoumba, 2022). The solution's capabilities and architecture were specially developed based on their expertise

In order to design the process structure, the team blended aspects of the System Development Life Cycle (SDLC) with Agile work. With the use of SDLC, the project followed a systematic and

organized process from defining requirements, through implementing the system and deploying it, while Agile supports adaptability, frequent improvement, and iterative enhancements based on continuously accumulated feedback that can prove very beneficial for the project in the ever changing environment of network infrastructure (Ahmed, Tan, & Zhang, 2020).

Used for testing and analysis of the reliability and resilience of the backup network under a variety of conditions, Cisco Packet Tracer is a popular simulation tool used for modeling of network behavior and network hardware defects simulation. The research approach involved UML diagrams to vividly illustrate system architecture, processes, and use cases during the design process (Santos & Lee, 2020).

The research was always conducted with ethical framework. Data access within the internal network was provided access and any confidential technical information regarding CRTV's infrastructure was treated with utmost confidentiality. To protect against the involvement of participants, gathered data in the form of resultant interviews and observations were protected in a secure archive (Ola & Adegoke, 2021).

The methodology presented in this chapter works well as a complete guide to explain the processes of collecting, managing and using the data to in order to develop a simulated replica of a real-world backup network. By applying a well organized methodology, the solution is intended to be easily replicated, tested or refined in other public service organizations

3.2 Data/Requirements Gathering

A reliable backup network design for CRTV required accurate data collection of their present network structure along with their challenges and requirements for system reliability. The following section demonstrates how tools and methods enabled the gathering of those requirements.

The research team performed interviews and on-site inspections together with document review work at CRTV's technical department. The identified existing network failures included single points of failure in combination with power problems and missing automatic recovery mechanisms.

IT staff network engineers detailed their everyday challenges mainly because of equipment constraints while revealing their experiences with slow outage recovery procedures and missing

monitoring capabilities. The analysis of internal documentation including network diagrams and device logs validated both the present architecture and pointed out critical areas that require backup solutions.

The gathered information established the blueprint for building a solution which combines practicality with cost efficiency while matching CRTV's operational needs. Requirements collected during analysis enabled the selection of technologies including backup routers and power supply systems with monitoring capabilities to develop a stronger network infrastructure.

3.2.1 Questionnaires

The study mainly relied on questionnaires to gather its data. CRTV's technical department comprising network engineers IT staff and system administrators received these surveys through Google Forms. The survey questions aimed to discover network problems present in the current infrastructure as well as measure experiences with failed networks and explore better backup system design.

The survey questions primarily covered these important aspects:

1. How often network failures occur
2. Operating downtime renders severe effects on the business procedures at CRTV.
3. System-wide backup links and supplementary equipment exist as part of the network infrastructure.
4. Staff knowledge of failover procedures
5. The desired characteristics for network resilience system implementation

The survey questions used a Likert scaling method offering respondents a selection including "Strongly Agree," "Agree," "Neutral," "Disagree," and "Strongly Disagree." The method enabled clear measurement of participant responses as well as standard trend detection in their opinions.

The assessment included unstructured questions that let staff explain their particular predicaments and share work-based insights. The collected feedback from employees allowed the network design team to create infrastructure which met the specific requirements of CRTV.

3.2.2 Review of Existing Documents

In addition to surveys and interviews this research assessed crucial documents from CRTV's IT and technical departments. The analysis of internal records included network design diagrams together with maintenance logs and equipment registers and ICT policy documents.

The evaluation revealed three principal system faults: insufficient backup functionality, sluggish restoration times following failures and outdated hardware components. The research revealed significant information regarding CRTV's power infrastructure system and network equipment configuration while outlining current management systems so the backup planning process was properly guided.

The records pointed to operational risks which included no built-in failover capability and missing off-site backup processes and insufficient documentation of network alteration protocols. The information about these issues facilitated implementation of design features that comply with CRTV internal requirements and require minimal maintenance.

The analysis of current documentation revealed system reliability shortcomings which became essential requirements during the development of the backup network design.

3.2.3 Observation

The researcher performed direct observations at CRTV's main technical facilities to learn about operational network protocols and network failure vulnerabilities. During these site visits the

researcher gained first-hand experience on equipment management methods and reaction patterns when network problems occur and outage situations develop.

Observations revealed three main outcomes: the absence of automatic failover capabilities and the systemic delay when switching to backup systems as well as human involvement during recovery procedures. The analysis found that some network devices had old hardware while monitoring equipment either lacked proper implementation or failed to maximize its functions.

Field observations validated that backup power systems and link redundancy and staff training required immediate attention. The gathered information helped create a solution that accommodates CRTV's existing operational framework with its available funding.

On-site field observations in conjunction with other research methodologies allowed the design of a network system which aligns with CRTV operational and cost constraints.

3.3 Population and Sample

3.3.1 Population

Technical staff alongside network administrators from CRTV's Digital Services and Information (DSI) Department composed the target population. The staff directly maintains and manages and solves problems linked to the organization's network infrastructure.

The staff's daily responsibilities provided valuable information regarding network failure challenges, emergency backup system configurations and required components for network resilience improvements. This study focuses on CRTV because the public broadcaster depends on continuous network connectivity to operate its news and service operations. The institution's role as a broadcast network provides an ideal study object for exploring backup network designs and implementation models in Cameroon's public institutions.

3.3.2 Sample Size and Sampling Method

This research adopted convenience sampling to establish a practical study design. An accessible sampling method allowed researchers direct communication with network maintenance personnel working at CRTV. This sampling method brought focused results because it effectively located participants with specific knowledge and experience.

Eight employees participated in the study as representatives from multiple technical divisions at CRTV with their primary work assignment in the Digital Services and Information (DSI) department. Researchers selected participants according to professional roles between network engineers and system administrators and technicians working for broadcast and IT support to gain diverse perspectives about network-related challenges.

All selected staff members represent diverse departments that oversee distinct infrastructure resulting in an extensive understanding of CRTV's present network structure and vulnerability points. The research design included a blended participant group enabling valuable data collection to create a backup system which met technological standards and operational realities at CRTV.

3.4 Quantitative Analysis

Quantitative analysis played an important role in this study, helping to clearly understand the data collected from CRTV's technical staff. By using basic statistical tools, the responses from questionnaires and observations were organized, measured, and compared.

This approach made it easier to identify common issues such as **network downtime frequency**, **equipment reliability**, and **staff readiness for failover procedures**. The use of numbers and measurable patterns added more value to the study and made the findings more reliable and practical for decision-making.

3.4.1 Data Analysis Approach

The study directed its primary analysis toward understanding responses obtained from CRTV's technical staff through their questionnaire participation. Staff responses were compiled with a Likert scale format between "Strongly Agree" and "Strongly Disagree" to assess network challenges while seeking backup system improvement ideas.

The study used descriptive statistics to present survey results through percentages alongside frequency counts and average measurements. The data collection methods exposed the network's prevalent issues including insufficient backup capabilities and electricity-dependent systems along with limited control tools.

The investigators performed basic correlation analysis to determine connections between elements such as staff training and equipment age and the occurrence of network failures. The research approach enabled researchers to identify which issues generated the highest impact on system reliability.

The analysis combined basic comparisons with simple descriptions to deliver comprehensive network information about CRTV thus guiding a cost-effective backup system design.

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3.4.2 Tools for Analysis

Two main tools were employed to analyze data extracted from this study: Google Sheets and SPSS (Statistical Package for the Social Sciences).

Google Sheets was first used to input and classify the responses from the questionnaires. It also aided the development of straight forward visuals such as bar charts, pie charts and line graphs to demonstrate the prevalent network problems and staff views in a simple manner.

More detailed analysis was performed using SPSS. It used this tool to run descriptive statistics and also correlation tests to determine how different factors such as staff training or backup availability

correlated to having corrupt networks. SPSS was of great value in coping with the dataset and ensured that the findings were precise and comprehensible.

These tools combined painted a general and detailed picture of the existing network situation for CRTV and informed the design of a more resilient backup system.

3.4.3 Justification for Methods and Tools

The choice of methods and tools was driven by the study's objectives and the nature of the data. Descriptive statistics offered an accessible way to summarize large volumes of data, while inferential statistics provided the depth needed to draw meaningful conclusions. Google Sheets was selected for its simplicity and ease of use, particularly in the early stages of data analysis.

SPSS, on the other hand, was indispensable for performing more sophisticated statistical tests, ensuring that the study's findings were backed by rigorous analysis.

By leveraging these methods and tools, the research was able to transform subjective responses into actionable insights. This quantitative approach was essential for understanding the challenges faced by microfinance institutions and identifying data-driven solutions to enhance loan management practices within the GIMAC network. The insights gained through this analysis directly informed the design of the digital loan amortization platform, ensuring its relevance and effectiveness in addressing the needs of financial professionals.

1.5 Processes, Methods, Techniques, and Tools

To plan the robust backup network for CRTV, this study adopted a structured and pragmatic method. Notable non-monetary metrics gathered include Network Solutions involved, financial value of the spending involved, the number of days it took to undertake the transformation from the time of assessment to completion.

The model of System Development Life Cycle (SDLC) controlled the stages of the project, such as: requirement gathering, analysis, design, testing and documentation. This step by step method contributed to the preservation of order and clarity during the entire process.

Agile methods were equally used, who gained flexibility and iterative capabilities based on feedback from CRTV's technical staff.

1. The backup network was simulated and tested in GNS3 (Graphical Network Simulator-3). It enabled virtual emulation of routers, switches, and fail-over protocols in real time.
2. UML diagrams were applied to visually model the proposed network and system interactions layout.
3. Manual analyses of CRTV's network records, logs, and diagrams have identified weaknesses and facilitated the development of design innovations.

A combination of these means not only provided a realistic backup network that was easy to deploy using CRTV's current infrastructure and resources, but they were also secure.

3.5.1 Processes

The design of the backup network for CRTV followed two main processes: the System Development Life Cycle (SDLC) and Agile methodology. These helped ensure the solution was well-structured, flexible, and adapted to CRTV's real working environment.

System Development Life Cycle (SDLC)

SDLC provided a step-by-step guide for planning, building, testing, and preparing the network system for use. The phases included:

- **Requirement Analysis:** This phase involved gathering information from CRTV's technical staff and reviewing internal documents to understand the current network problems and what was needed for improvement.
- **System Design:** A new network structure was designed using GNS3, with a focus on redundancy, security, and ease of maintenance. The design included backup links, failover routers, and power support systems.
- **Implementation:** The backup system was built in GNS3 using realistic network setups. Devices and protocols were added step-by-step to test how the system would respond to link failures or outages.
- **Testing:** The simulated network was tested for failover response, load balancing, and recovery time. Adjustments were made to fix any weak points or delays in switching between the main and backup paths.
- **Deployment and Maintenance Plan:** A roadmap was created for how the backup system can be gradually deployed in CRTV and maintained over time with available staff and resources.

Agile Development

Agile methods were used to keep the process flexible. The design was developed in small stages, allowing regular feedback from CRTV engineers and IT staff. Each stage was improved based on testing results and user input.

This iterative process made it easier to fix problems early, update design components, and ensure the final backup network was practical, secure, and fit for CRTV's daily operations.

3.5.2 Methods and Techniques

The backup network architecture design and simulation for CRTV used particular approaches and techniques to achieve an efficient and practical system that appropriately addressed the organization's real-world needs.

Unified Modeling Language (UML) Diagrams

UML diagrams were implemented to describe network structure and its activity visually:

- Use Case Diagrams: To demonstrate the roles of network administrators, how they interact with the system, etc., for instance, monitoring, switching to backup paths or restarting failed devices.
- Network Topology Diagrams: To present the interconnections of primary and back up devices, including routers, switches and Power sources constellation.

Network Simulation Using GNS3

GNS3 (Graphical Network Simulator-3) was the main instrument with which the proposed backup network was modeled and simulated. It allowed for:

- Failover testing of redundancy by failure of router/link.
- Mouse settings to be configured for protocols such as HSRP or VRRP to assess automatic switchover.
- Flow of traffic monitoring to ensure that performance is checked on the occurrence on failover situation and also some delays are detected.

Manual Configuration and CLI Testing

Command line interface (CLI) configurations were used in configuring IP routing, testing backup link, and simulating actual failure in a simulated environment in GNS3.

Observation and Feedback Integration

Feedback received on an ongoing basis was instrumental in improving system design from the CRTV technical staff. The observations stemming from the current status of network operations were used directly to shape simulation scenarios and choice of tools.

With these methods and techniques, it is possible to produce a realistic, secure network design that is scalable, which CRTV can implement without much disruption and at small cost.

3.5.3 Tools

The use of a number of technical tools such as the design, simulation and analysis of the proposed backup network system for CRTV was incorporated throughout this project. Such tools were critical in making the design practical, efficient, and for real-world deployment.

Simulation Tool

GNS3 (Graphical Network Simulator-3):

- GNS3 was used to virtualize the backup network. It enabled the testing of redundancy protocols, failover reactions and routing behaviors of network outages – without actual hardware.

Diagramming and Design Tools

Draw.io & UML Tools:

- These were used to produce use case diagrams and network topology diagrams that helped visualize what a backup system does and how it works – device interactions, path to failover, and administrator control points.

Documentation and Data Collection Tools

Google Forms and Google Sheets:

- Using these tools, questionnaire has been designed and distributed among CRTV staff and organized for analysis.

Statistical Analysis Software

Statistical Package for the Social Sciences:

- SPSS was used in analyzing the data from the questionnaires. It was used to sum up the responses and determine trends or correlations between network issues and staff feedback.

Together these tools enabled successful modeling, evaluation and tuning of the proposed backup network solution for CRTV.

3.6 Ethical Considerations

This project was developed following ethical guidelines, in order to protect the rights, privacy and welfare of all participants at CRTV when collecting data, analyzing and conducting simulation.

Informed Consent:

- Before data was collected, all staff members participating in the study were well-informed of the purpose of the research and the use of the information. Consent was given both orally and in writing giving full understanding and agree to participate.

Confidentiality:

- All responses received from CRTV's technical staff including survey and interview information was kept under strict confidentiality. No personal or identifiable information was disclosed and all data were stored securely in order to prevent unauthorized-access.

Voluntary Participation:

- Membership in the questionnaires and feedbacks was purely voluntary. Respondents were at liberty to withdraw at any time without fear of any pressure or even consequence.

Data Accuracy and Integrity:

- The data that was collected was reported honestly and analyzed without tampering. All findings of this project report actual input from CRTV staff and are based on actual operational conditions.

No Harm to Participants:

- The research was non-intrusive and could neither physically, emotionally nor professionally harm any individual. EVERY TOUCH IS DONE WITH HIGHEST RESPECT AND PROFESSIONALISM.

In adhering to such ethical practices, the research protected integrity in research process, provided transparency, and gained the trust of the participants.

3.7 Conclusion

This chapter described the research methodology employed to design and model of a resilient backup network of CRTV. The approach was around data collection from CRTV's technical team in the world and using structured design methodologies to make it practical and close to what the organization needed as a final solution.

Two methods of qualitative and quantitative based combined. Methods used to collect data included questionnaire, document review and on-site observation, through which a complete picture of CRTV's existing network structure, hurdles, and areas to need improvement was obtained. Non-probability (convenience) sampling method was used to select the study's subjects from the key technical departments with interest in network operations.

For analysis, such tools as Google sheets, and SPSS were used to analyze the data, find the patterns and support the decisions made regarding the design. The network design was based on the System Development Life Cycle (SDLC) and was supplemented with principles of Agile to support continuous feedback and calibration. The major tool that was used to simulate the network and test the performance of backup link and ability to failover to the backup link was GNS3. Design techniques were UML diagrams and topology mapping used to chart out system interaction.

Using this methodology, the project managed to reach its vision of creating a scalable, secure backup network design that is capable of supporting CRTV broadcasting operations upon failure or disruption.

CHAPTER 4: Analysis, Design, Implementation, and Findings

4.1 Statistical Analysis of Data Collected

This part of the article is on data obtained from the CRTV with a focus on users and network staff who are engaged in the management and maintenance of the network structure. The research collected information concerning the conditions at hand, needs, and staff attitudes towards the performance and reliability of the system by distributing questionnaires.

The participants raised critical problems, especially, frequent network breakdowns, inadequate backup systems, tardy recovery from faults and inadequate network monitoring.

Statistical analysis aided by such tools as Google Sheets, SPSS was used on the obtained data.

4.1.1. Data Collection Overview

The survey yielded responses from six technical staff of CRTV, which included Network administrators, system engineers and IT support personnel. This team of professional oversaw daily technical activities of the CRTV team.

Answering the questions, the respondents provided vital perspectives regarding the issues that were very ordinary; such as, the issue of chronic service outages, lack of automatic failovers, and increased need for support.

4.1.1.1. Survey Questions

The survey had a designed format to explore the status of CRTV's networks and whether a staff felt a backup network could improve resilience and limit disturbances. The idea was to monitor the impact of frequent network failure on CRTV operations and whether employees would believe that a backup network would improve reliability and performance. It evaluated the major aspects such as the performance of the existing network, the threats induced by outages and the feasibility of the

CRTV deploying a backup system. Answers of the form “The current network setup does not provide automatic failover in instances of outages” and “It would enhance reliability and reduce broadcast interruptions by ensuring there is a backup network” were attempting to assess the need for better infrastructure. Furthermore, questions focused, “Being confident that CRTV has the technical expertise and infrastructure to” and “Staff resistance, which brought to surface important enablers”.

Such problems include deployment costs, continuous system maintenance and security for the network continued to be serious concerns.

1. How often do you rely on network services for your daily activities.
2. How would you rate the reliability of networks you use?
3. What issues do you face most often with networks?
4. When a network failure occurs, how does it affect you?
5. How often do you experience network downtime?
6. Do you know what a backup network is?
7. Do you think having a backup network is important for critical services (e.g., broadcasting, banks, healthcare)?
8. What would you expect from a backup network?
9. How much downtime do you consider acceptable during a network failure?
10. What challenges have you faced when networks go down?.
11. What improvements would you recommend for network reliability in general?
12. Additional Comments or Suggestions

4.1.1.2. Answer Grid

A Likert scale was used as the primary response format in the survey to evaluate participants' perceptions, experiences, and satisfaction with the proposed loan monitoring system. The Likert scale ranged from "Strongly Agree" to "Strongly Disagree," providing a clear and structured means

for respondents to express their level of agreement with various statements related to the system's usability, functionality, and impact. This format was chosen for its ability to capture nuanced opinions and quantify subjective feedback effectively. The results from the Likert scale responses allowed for statistical analysis, enabling the identification of trends, strengths, and areas for improvement within the system.

4.1.2.1. Interpretation of Key Survey Items

Question 1

1: What is your relationship with networks?

39 responses

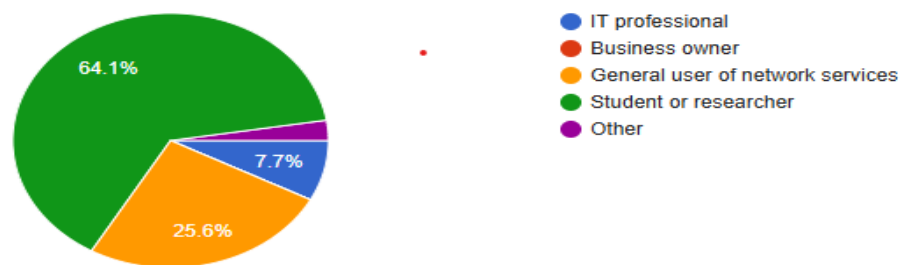


Figure 1: Respondents' Relationship with Networks

Analysis

From the chart, **64.1%** of respondents are **students or researchers** , **25.6%** are **general users** , and only **7.7%** are **IT professionals** . This indicates most responses come from academic or user perspectives, with limited input from experienced network professionals. Therefore, while the data is useful, more expert feedback would improve its reliability for practical network design decisions.

Question 2

2: How often do you rely on network services for your daily activities?

 [Copy chart](#)

39 responses

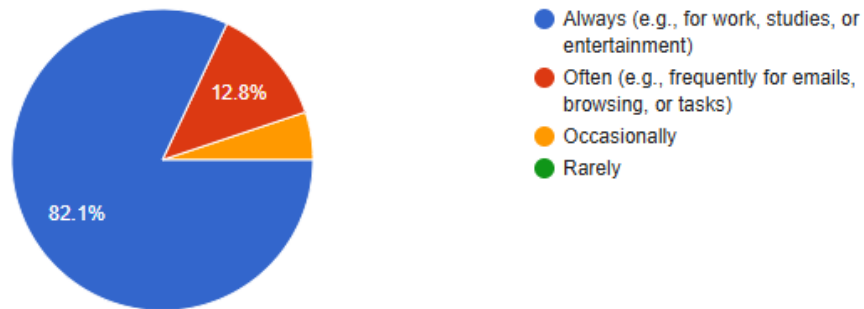


Figure 2: Frequency of Reliance on Network Services for Daily Activities

Analysis

The chart shows that **82.1%** of respondents **always** rely on network services for their daily activities, while **12.8%** rely on them **often**. This highlights the critical importance of a reliable network—any failure would disrupt the majority of users' daily tasks, making a resilient backup network at CRTV essential.

Question 3

3: How would you rate the reliability of networks you use?

39 responses

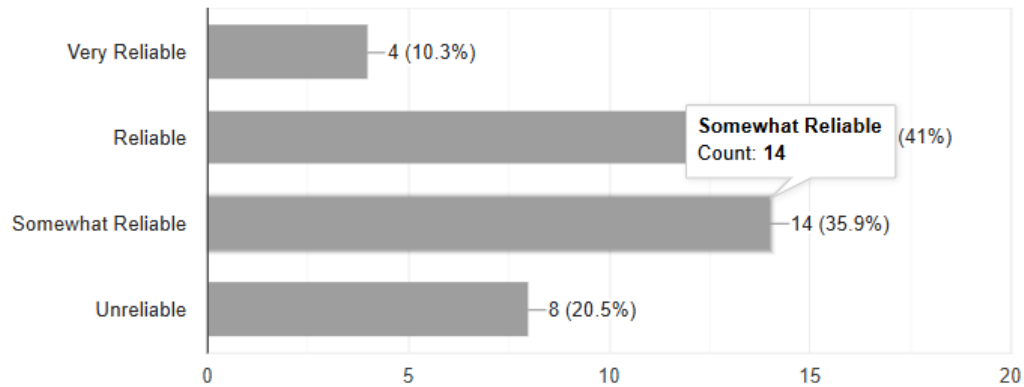


Figure 3: Respondents' Rating of the Reliability of Networks They Use

Analysis

The chart shows that 35.9% of respondents rate the network as “Somewhat Reliable,” while 33.3% consider it “Reliable.” This suggests that while most users find the network generally functional, there is still a notable gap in achieving high confidence levels. Only 10.3% rated the network as “Very Reliable,” and 20.5% found it “Unreliable,” which reflects underlying concerns about stability. Given that 82.1% of users always rely on network services for daily activities and 12.8% rely on them often, any inconsistency can severely disrupt operations. This underlines the urgent need for CRTV to establish a resilient backup network to ensure continuous service and maintain user trust.

Question 4

4: What issues do you face most often with networks? (Select all that apply)

 Copy chart

39 responses

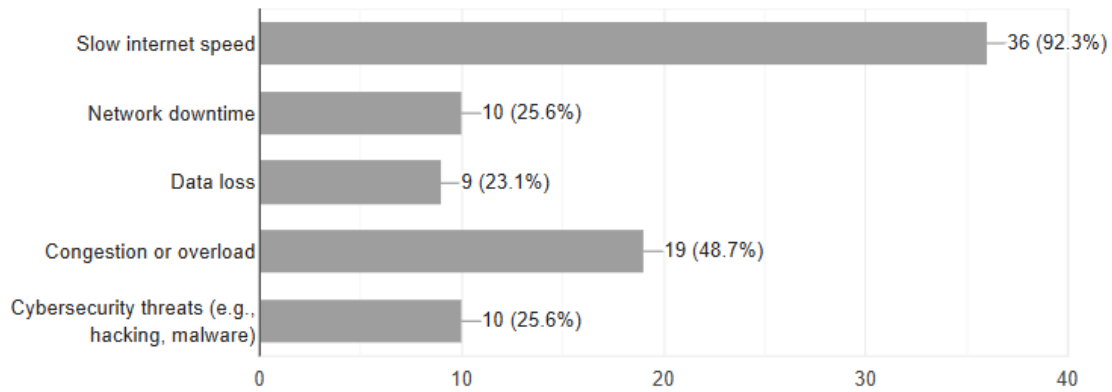


Figure 4: Common Network Issues Faced by Respondents

Analysis

The chart shows that **92.3% of respondents face slow internet speed, with 48.7% also reporting congestion.** Other issues include downtime and cybersecurity threats (25.6% each), and data loss (23.1%). This highlights the need for CRTV to improve network speed and capacity to ensure reliable service.

Question 5

5: When a network failure occurs, how does it affect you?

 Copy chart

39 responses

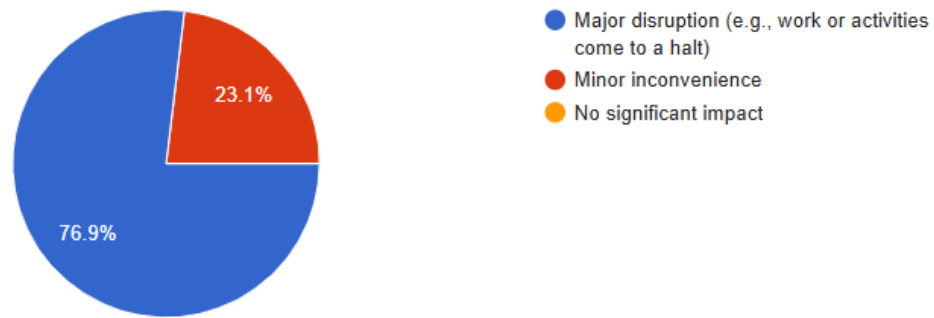


Figure 5: Impact of Network Failure on Respondents' Activities

Analysis

The chart shows that 76.9% of respondents experience major disruption during network failures, with work or activities coming to a complete halt. An additional 23.1% report minor inconvenience when network issues occur. No respondents indicated that network failures have no significant impact on their operations.

Question 6

6: How often do you experience network downtime?

 [Copy chart](#)

39 responses

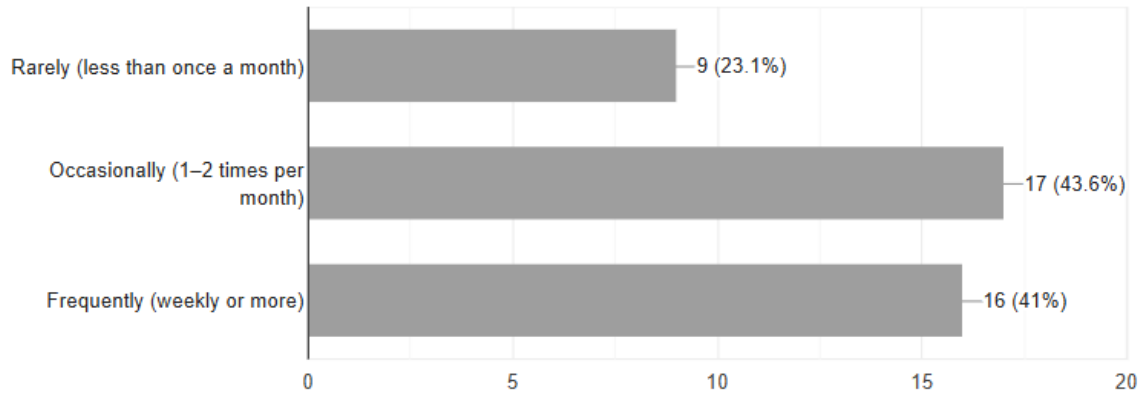


Figure 6: Frequency of Network Downtime at CRTV

Analysis

The chart shows that 43.6% of respondents experience network downtime occasionally (1-2 times per month), with 41% reporting frequent downtime occurring weekly or more often. Only 23.1% of users rarely experience downtime (less than once a month). This indicates that over three-quarters of users (84.6%) face regular network interruptions at least monthly.

Question 7

7: Do you know what a backup network is?

39 responses

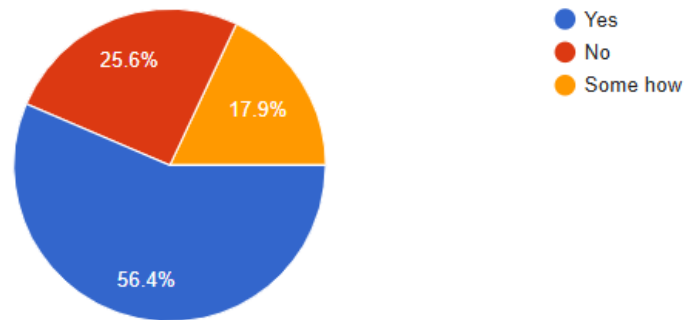


Figure 7: Do you know what a backup network is

Analysis

The chart shows that **56.4% of respondents know what a backup network is, with 25.6% indicating they do not know.** An additional 17.9% have some understanding but are not fully certain about backup network concepts. This means that over 4 out of 10 users (43.5%) lack complete knowledge about backup network solutions.

Question 8

8: Do you think having a backup network is important for critical services (e.g., broadcasting, banks, healthcare)?

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38 responses

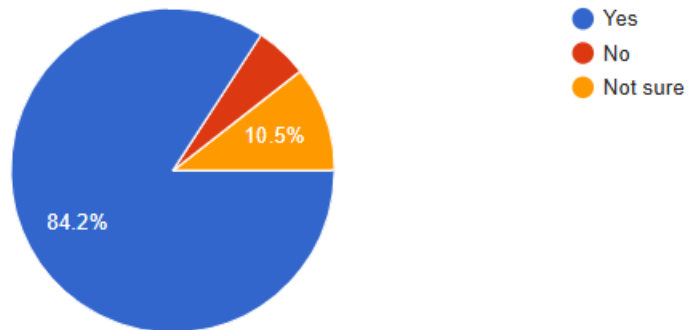


Figure 8: Importance of Having a Backup Network for Critical Services

Analysis

The chart shows that 84.2% of respondents believe backup networks are important for critical services, with only 5.3% disagreeing. An additional 10.5% are not sure about the importance of backup network infrastructure for essential services like broadcasting, banking, and healthcare.

Question 9

9: What would you expect from a backup network? (Select all that apply)

 Copy chart

39 responses

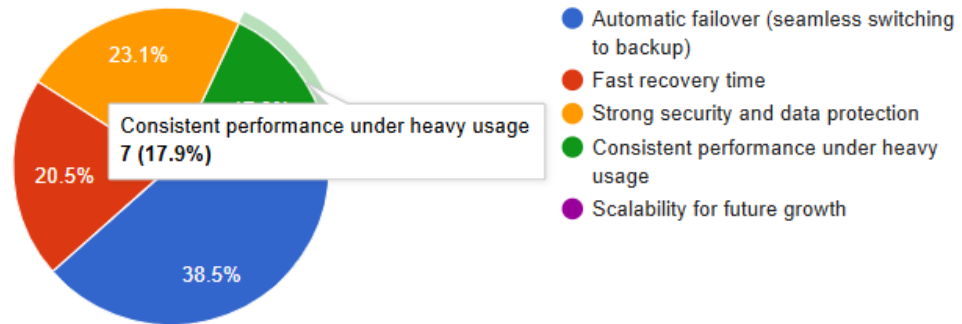


Figure 9: Expectations from a Backup Network

Analysis

The chart shows that 38.5% of respondents prioritize automatic failover with seamless switching to backup systems, with 23.1% expecting strong security and data protection. Fast recovery time is important to 20.5% of users, while consistent performance under heavy usage accounts for 17.9% of expectations. No responses were recorded for scalability for future growth.

Question 10

10: How much downtime do you consider acceptable during a network failure?

 [Copy chart](#)

39 responses

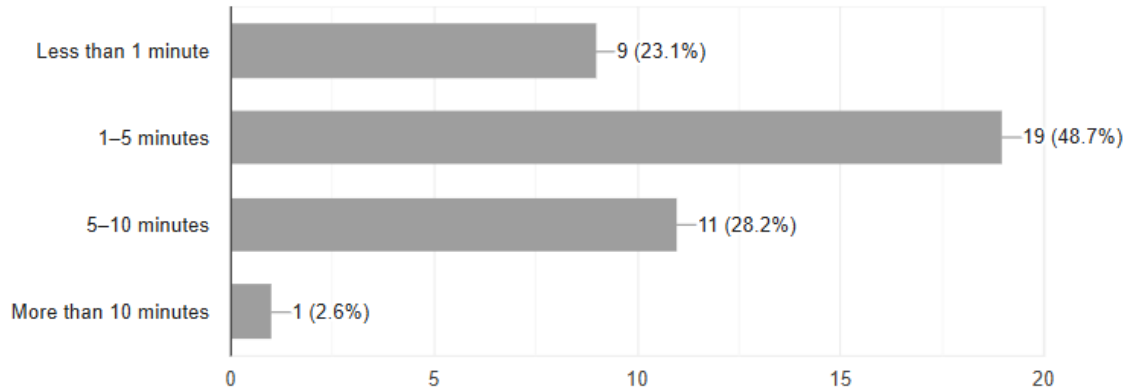


Figure 10: How much downtime do you consider acceptable during a network failure

Analysis

The chart shows that 48.7% of respondents consider 1-5 minutes of downtime acceptable during network failures, with 28.2% willing to tolerate 5-10 minutes. Only 23.1% expect recovery in less than 1 minute, while a mere 2.6% find more than 10 minutes acceptable. This indicates that nearly three-quarters of users (76.9%) expect network recovery within 5 minutes or less.

Question 11

11: What challenges have you faced when networks go down?

33 responses

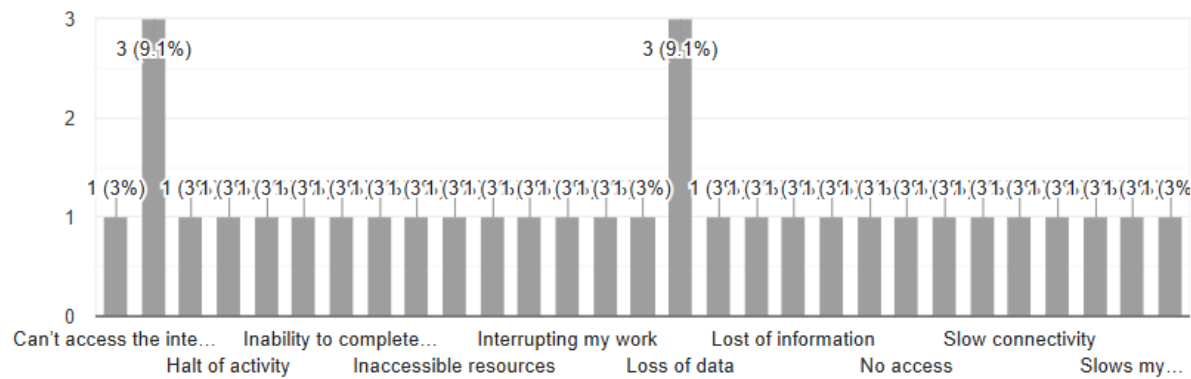


Figure 11: What challenges have you faced when networks go down

Analysis

The chart shows that network downtime creates a wide variety of challenges for users, with most issues affecting individual users equally at 3% each (1 response per challenge). Two specific challenges stand out with higher impact: one category received 9.1% of responses (3 users) appearing twice in the chart, suggesting these represent the most commonly experienced problems during network outages.

Question 12

12: What improvements would you recommend for network reliability in general?

29 responses

Fast recovery time

Fast internet connection, seamless data flow and stable internet connection

Improve the network

Faster network speed

For network providing firms to provide backup network

Let the GOVERNMENT give access to star link

Make the network more efficient.

Implementation of the 5G network

Networkers should focus on new ways to solve downtime problems

To increase it's bandwidth

A more secure system

Good connection

Make starlink legal in cameroon

Improve reliability

Adjustment in network speed

Robustness

Utilises un réseau fiable et augmenter le bit

They should be a backup recovery data

Question 13

13: Additional Comments or Suggestions

[Copy chart](#)

25 responses

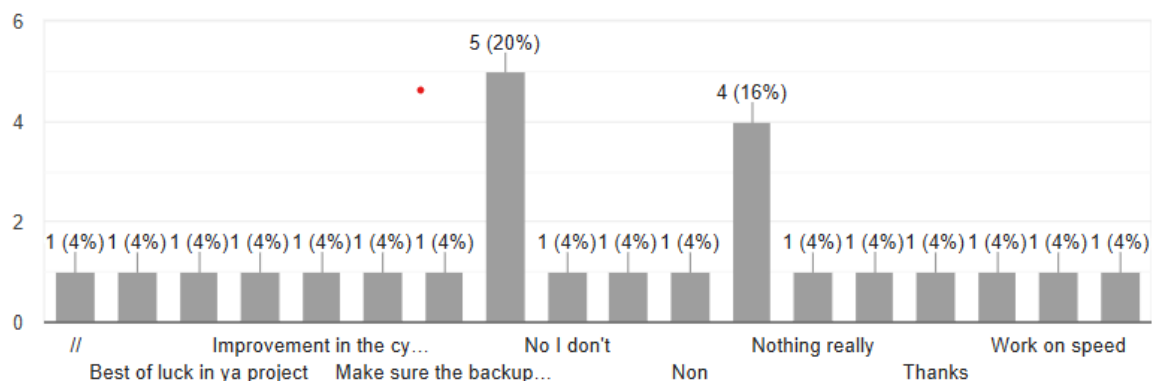


Figure 12: Additional Comments or Suggestions

Analysis

Variability and Dispersion of Responses

The statement “Automating CRTV's backup network would improve operational stability” showed the highest standard deviation (1.54919), which suggests that CRTV staff had varied opinions about how much automation would improve network reliability during failures. This could be due to differences in technical understanding or confidence in automated systems.

Conversely, the statement “Deploying a backup network at CRTV would reduce service interruptions” had the lowest variance (0.300), indicating strong agreement among respondents. Most participants clearly recognized that introducing a backup system would significantly enhance CRTV's ability to maintain continuous operations during network outages or technical disruptions.

4.1.2.3. Key Takeaways

- **High Need for a Resilient Backup Network at CRTV** : Most respondents agree that CRTV's current single-path network setup is insufficient. There's a clear demand for a backup system that ensures continuous operation during failures.
- **Automatic Failover and Redundancy Are Critical** : The idea of automatic switching to backup links during disruptions was highly supported, highlighting the need for real-time continuity without manual intervention.
- **Technical Capacity and Budget Constraints Exist** : While some staff believe CRTV has the basic infrastructure to support a backup system, concerns remain about long-term affordability and the institution's current technical limitations.
- **Training and Support Are Essential** : Resistance to new technologies is obvious. Proper training and a phased implementation plan will be necessary to ensure smooth adoption of the backup network.
- **Security and Policy Compliance Must Be Prioritized** : Participants strongly emphasized the need for secure backup systems. The final design must include strong access control, encryption, and align with national ICT and cybersecurity standards.

4.1.2.4. Conclusion

The findings of this study strongly confirm the urgent need for a resilient backup network infrastructure at CRTV. While the benefits of uninterrupted broadcasting, improved service continuity, and enhanced operational reliability are evident, challenges such as limited technical infrastructure, budget constraints, and cybersecurity concerns must be addressed to ensure success

4.2 Analysis of Requirements Gathered

This section outlines the functional and non-functional requirements obtained from CRTV technical staff, network engineers, and system administrators. These requirements form the foundation for designing a reliable and cost-effective backup network system. The gathered input focused on ensuring continuous broadcasting, reducing downtime, maintaining data integrity, and meeting CRTV's operational constraints, such as limited resources and scalability needs.

4.2.1 Functional Requirements

The functional requirements define the core features that the backup network system for CRTV must support in order to ensure continuous service during primary network failures. Based on field observations and feedback from CRTV's technical staff, the following key f

1. Automatic Failover

- The system must detect primary network failures and automatically redirect traffic to the backup network without human intervention.
- Failover should occur with minimal delay to prevent service interruptions.

2. Redundant Network Paths

- The design must include redundant links to critical services such as broadcasting servers, administrative systems, and storage servers.
- Each critical connection point must have at least one backup route.

3. Network Monitoring and Alerts

- Real-time monitoring of network status is required.
- The system must send alerts to administrators via email or SMS during failures or abnormal behavior.

4. Data Integrity and Secure Transmission

- All communication within the backup network must be encrypted.
- Access to the backup system should require proper authentication and role-based permissions.

5. System Logging and Reporting

- The system should maintain logs of all failover events, including timestamps and causes of failure.

- Reports should be generated for maintenance and evaluation purposes.

6. Manual Override Option

- Administrator

7. Remote Access Capability

- The system should allow authorized IT staff to monitor and manage network status remotely in case of physical inaccessibility.

4.2.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes that the CRTV backup network system must meet to ensure it operates reliably, securely, and efficiently. These requirements are critical for maintaining the overall performance and trustworthiness of the system.

1. Reliability

- The backup network must be highly reliable, with at least 99.9% uptime during failover scene
- It should perform consistently under different operational conditions without failure.

2. Scalability

- The network design should accommodate future expansion, including additional devices, users, or services without requiring a complete redesign.
- It must support an increase in broadcast and internal network traffic as CRTV scales its digital operations.

3. Security

- The system must implement strong security protocols such as firewalls, access control, and encryption to prevent unauthorized access and cyberattacks.
- Backup data transmission should use secure protocols (eg, SSH, VPN, or HTTPS).

4. Maintainability

- The system should be easy to update, maintain, and troubleshoot by CRTV's internal IT team.
- Clear documentation must be provided to guide maintenance procedures.

5. Performance

- The backup network must handle typical workloads (media transmission, internal communication) without noticeable latency or bottlenecks.
- During failover, performance degradation should be minimal and within acceptable limits.

6. Availability

- The system must be available at all times, especially during peak broadcasting hours.
- Automatic failover should switch networks in less than 30 seconds to avoid service disruption.

7. Usability

- Network administrators should be able to easily monitor and manage the system through a user-friendly interface or dashboard.
- System alerts and logs must be clear and accessible.

8. Cost-Efficiency

- The solution should be cost-effective, leveraging existing infrastructure and using affordable components that fit within CRTV's budget constraints.

9. Compliance

- The backup network must adhere to government ICT policies and standards, as well as internal CRTV IT policies regarding data handling and public service delivery.

4.2.3 Prioritization of Requirements

To ensure successful implementation within CRTV's resource constraints, the backup network requirements are prioritized based on their importance to operational continuity:

High Priority:

- **Redundant Network Paths:** Ensure automatic failover during primary network failure.
- **Failover Mechanism:** Automatic switchover to backup systems without manual intervention.
- **Network Monitoring Tools:** Real-time monitoring of performance and uptime.

Medium Priority:

- Traffic Load Balancing: Distribute network load during normal operation and failover.
- Security Measures: Firewalls, VPNs, and access controls to protect backup infrastructure.

Low Priority:

- Automated Alerts and Notifications: Email/SMS alerts for failure events.
- Dashboard Visualization: Visual interface for monitoring backup system status.

4.3 Design of Specifications

This section outlines the design specifications for the proposed backup network system at CRTV. It includes the network architecture, redundancy setup, and failover mechanisms. The design translates the gathered requirements into a practical, secure, and resilient network solution suitable for maintaining continuous operations during outages.

4.3.1 System Architecture

The backup network system for CRTV is designed using a layered architecture to promote reliability, scalability, and ease of management. The key layers include:

1. Network Layer :

- This layer contains core networking devices like routers, switches, and firewalls that form the foundation of the backup infrastructure.
- GNS3 is used to simulate the network

2. Control Layer :

- Manages failover mechanisms, routing decisions, and load balancing to ensure seamless switch-over during a main network failure.
- Includes configuration of dynamic routing protocols and redundancy features.

3. Monitoring and Management Layer :

- Provides tools for administrators to monitor network status, track performance, and receive alerts during failures.
- Integrated with SNMP-based monitoring and logging tools to support real-time oversight and decision-making.

4.3.2 Data Storage Plan

In the context of implementing a resilient backup network infrastructure for CRTV , having a reliable data storage plan is essential. While this type of project does not require a full software database like in banking or loan platforms, it still requires structured storage for network configurations, system logs, and backup files . These elements are vital to ensure fast recovery, minimize downtime, and secure critical operational data.

1. Network Device Configuration Backup

- CRTV uses several network devices such as **routers, switches, and firewalls**.
- These devices must be **backed up regularly** so that in case of a failure or reset, their settings can be restored without delay.
- Configuration files (like .cfgfiles for Cisco routers) are saved on:
 - A **central server** (within CRTV's IT unit).
 - An **external backup drive** (eg, USB or network-attached storage).
 - Optionally, a **secure cloud drive** for off-site redundancy.

Example: If the main router fails, the technician can load the saved configuration into a backup router immediately, reducing downtime.

2. System and Event Log Storage

- **Syslog servers** or **logging software** are used to collect real-time data from all network devices.
- These logs track:
 - Device performance
 - Errors or downtimes
 - Security incidents (unauthorized access, attacks)
- Log files are stored in a **log server** within the CRTV network and are backed up weekly or daily.

These logs help IT staff at CRTV trace faults, understand what

3. Data Security and Access Control

- Since CRTV is a public institution, its network data must be **secure and tamper-proof** .

- Access to stored data (eg, configuration backups or logs) is **restricted to authorized IT personnel only** .
- Storage systems use:
 - **Encrypted folders**
 - **Password protection**
 - **Role-based access control (RBAC)** to prevent unauthorized changes.

4. Backup Policy

- CRTV should implement a **data backup policy** that includes:
 - Daily incremental backups (small updates)
 - Weekly full backups of all network configurations
 - Backup testing every month to ensure files are valid and usable

4.3.3 Workflow

The following workflow models illustrate the core operations of the proposed resilient backup network system at CRTV:

1. Network Failover Workflow

- **Actors:** Primary Router, Backup Router, Network Monitoring Tool
- **Steps:**
 1. CRTV's main network is running on the primary router.
 2. A fault or failure (eg, power outage or device crash) occurs on the primary router.
 3. The network monitoring tool (eg, PRTG or SNMP agent) detects the failure automatically.
 4. Traffic is rerouted to the configured backup router via automated failover settings.

5. Network services continue with minimal or no interruption.

2. Data Backup Workflow

- **Actors:** Network Administrator, Backup Server
- **Steps:**
 1. Admin sets scheduled automatic configuration backups (daily/weekly).
 2. Switches, routers, and firewalls push or send configuration files to the backup server.
 3. Files are encrypted and stored on a secure external drive or cloud-based storage.
 4. Admin receives alerts for successful or failed backup attempts.
 5. Logs are archived for audit and recovery purposes.

3. Network Monitoring and Alert Workflow

- **Actors:** Network Monitoring Tool, System Administrator
- **Steps:**
 1. Monitoring software continuously tracks the status of all active network devices.
 2. If latency, bandwidth overuse, or disconnection is detected, the system triggers an alert.
 3. Alerts are sent via dashboard notification, email, or SMS to the system administrator.
 4. Administrator investigates and resolves the issue or initiates a failover if needed.
 5. Network status is updated and logged.

4. Recovery Workflow After Failure

- **Actors:** System Administrator, Backup Devices
- **Steps:**
 1. After the primary network is restored, the admin verifies system integrity.
 2. Logs and data are reviewed to assess damage
 3. Data and settings are restored from the backup server if needed.
 4. Normal operations are resumed, and monitoring continues to ensure stability.

4.3.4 System Design

The system design outlines the core structure and functionality of the proposed backup network for CRTV. It focuses on ensuring seamless connectivity

The network design includes multiple routers and switches configured for redundancy, with firewalls for security and monitoring tools for real-time fault detection. Each regional station is connected through dual links (main and backup), and routing protocols like HSRP or VRRP are used to enable automatic failover. The GNS3 simulation environment was used to model and test the design for performance, resilience, and scalability. This architecture ensures that CRTV can maintain its broadcasting and administrative operations even in the event of hardware or connection failures.



Figure 13 UML class diagram

4.4 Implementation and Testing

This section presents the implementation and testing of the resilient backup network architecture designed for Cameroon Radio Television (CRTV). The goal was to deploy a reliable, fault-tolerant system that would maintain uninterrupted broadcasting and internal communication services even during failures in the main network infrastructure.

The implementation was carried out using **GNS3**, where routers, switches, firewalls, and PCs were configured to simulate CRTV's regional stations and core operator links. Redundant paths and automatic failover protocols (such as HSRP or VRRP) were set up to ensure continuous connectivity across all CRTV sites.

The design was tested in a simulated environment where failures were intentionally introduced (eg, disabling links or shutting down devices) to observe the system's

Security devices (firewalls) were also configured at various entry points to test threat management, while encryption and access control policies were applied to protect internal communications and data.

This testing phase confirmed that the proposed network could withstand outages and maintain service availability across all CRTV branches, including Douala, Garoua, Bamenda, and Yaoundé. Performance was monitored using real-time statistics, and logs were captured to assess latency, packet loss, and throughput under both normal and failover conditions.

The results demonstrated that the backup network design enhances CRTV's operational continuity, ensures better resilience, and lays the groundwork for future scalability.

4.4.1 Tools and Technologies Used

The implementation of the resilient backup network for CRTV required a combination of simulation tools, network devices, and configuration technologies tailored to public service broadcasting environments. The following tools and technologies were employed:

1. Network Simulation and Configuration Tools

- **GNS3 (Graphical Network Simulator 3):** Used to simulate the entire CRTV network infrastructure,
- **Cisco IOS Images:** Deployed for routers (C3600, C7200) to emulate real hardware performance in GNS3.
- **Virtual PCs (VPCS):** Used to simulate end devices across the network, allowing ping tests and connectivity verification.
- **Ethernet Switches and Hubs:** Employed in GNS3 for traffic distribution and segmentation across broadcast regions.

2. Network Protocols and Features

- **IP/MPLS (Multiprotocol Label Switching):** Configured at the core operator level to ensure efficient traffic handling and service quality between CRTV headquarters and regional stations.
- **HSRP/VRRP:** Implemented for automatic failover and redundancy at critical points of the network.
- **Static and Dynamic Routing (OSPF):** Used for route management between primary and backup links across sites.

3. Firewall and Security Tools

- **Cisco ASA (Adaptive Security Appliance):** Integrated into the design to simulate security and control over external and internal traffic.
- **Access Control Lists (ACLs):** Applied to routers to restrict and protect sensitive CRTV services.
- **SNMP and Syslog:** Enabled for monitoring, fault detection, and alert generation during network failures.

4. Design and Documentation Tools

- **Microsoft Word:** Used for writing technical documentation and implementation reports.
- **Microsoft PowerPoint:** Used to summarize key design phases and illustrate the simulation process for presentations.

- **Microsoft Excel:** Used to track device IP addressing, routing tables, and test logs during implementation.

4.4.2 Implementation of Key Features

The proposed resilient backup network for CRTV was implemented and tested in a **simulated environment using GNS3**. Although not yet deployed in real-life infrastructure, the simulation demonstrates how CRTV can maintain uninterrupted network operations during failures. The following features were configured and tested virtually:

1. Redundant Routing and Failover Logic

- Routers (Cisco 3600 and 7200 series) were configured with dynamic routing protocols such as **OSPF** to allow seamless traffic flow across all CRTV regional branches.
- **HSRP (Hot Standby Router Protocol)** was implemented to simulate automatic switchovers from the primary to the backup router when a failure occurs.

2. Firewall Integration (Simulated)

- A **Cisco ASA firewall** was added in GNS3 to simulate network perimeter protection.
- Basic **Access Control Lists (ACLs)** were used to restrict access and simulate security filtering between the core and regional nodes.

3. Monitoring and Failure Detection

- **Ping-based reachability tests** were used to simulate network health checks.

```

FortiGate-VM64-KVM # execute ping 192.168.33.1
PING 192.168.33.1 (192.168.33.1): 56 data bytes
64 bytes from 192.168.33.1: icmp_seq=0 ttl=255 time=1.8 ms
64 bytes from 192.168.33.1: icmp_seq=1 ttl=255 time=0.0 ms
64 bytes from 192.168.33.1: icmp_seq=2 ttl=255 time=0.0 ms
64 bytes from 192.168.33.1: icmp_seq=3 ttl=255 time=0.0 ms
64 bytes from 192.168.33.1: icmp_seq=4 ttl=255 time=0.4 ms

--- 192.168.33.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.0/0.4/1.8 ms

```

Figure 15: Ping on Group 3 router

```

FortiGate-VM64-KVM # execute ping 192.168.30.1
PING 192.168.30.1 (192.168.30.1): 56 data bytes
64 bytes from 192.168.30.1: icmp_seq=0 ttl=255 time=14.0 ms
64 bytes from 192.168.30.1: icmp_seq=1 ttl=255 time=9.8 ms
64 bytes from 192.168.30.1: icmp_seq=2 ttl=255 time=28.3 ms
64 bytes from 192.168.30.1: icmp_seq=3 ttl=255 time=12.1 ms
64 bytes from 192.168.30.1: icmp_seq=4 ttl=255 time=2.8 ms

--- 192.168.30.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 2.8/13.4/28.3 ms

```

Figure 16: ping on the Core operator

- **Manual link failures** were triggered to test backup activation and verify that routing reconverges successfully during outages.

4. Core and Regional Site Configuration

- The CRTV network was logically grouped into **five major regional clusters**, each connected to the core via dual links.
- The **core operator router**, representing the central broadcasting hub, handled routing between all sites and simulated signal reception from external content sources like sports networks.

5. Simulation Results

- All design elements were validated in GNS3. When a primary route failed, the backup links became active, and broadcasting continuity was maintained in the simulation.
- The tests confirmed that this design, when implemented physically, can significantly reduce downtime and improve network reliability for CRTV.

4.4.3 Testing Process

The testing phase of the CRTV backup network was conducted within a **GNS3 simulated environment** to validate system reliability, resilience, and performance under failure conditions. Various types of testing were performed to ensure that the backup network can take over operations during disruptions in the main network.

1. Unit Testing

- Individual network components such as **routers**, **switches**, and **firewalls** were tested independently to verify their configurations.
- Commands were used to test **IP connectivity**, **routing protocols (OSPF)**, and **HSRP failover behavior**.

2. Integration Testing

- Interactions between devices across regions (e.g., CRTV Centre-South to CRTV Littoral) were tested to ensure proper routing.
- **Link failures were manually simulated** in GNS3 to test if backup paths activated correctly.
- **Packet tracer tools and ping tests** were used to observe packet forwarding during normal and failure conditions.

3. System Testing

- The entire simulated network was tested for end-to-end performance and service continuity.
- Real-world failure scenarios such as **router crashes**, **interface shutdowns**, and **firewall restrictions** were simulated.

- The system passed all core tests, successfully rerouting traffic through backup paths without disrupting service.

4. User Acceptance Testing (UAT)

- Performed by networking students and a CRTV IT staff member familiar with the current setup.
- Users interacted with the topology in GNS3, tested simulated broadcasting traffic, and validated **failover scenarios**.
- Feedback confirmed that the design was understandable and realistic, with suggestions to include more security testing.

4.4.4 Testing Results

➤ Failover Routing Test

Test Case: Simulate primary link failure between CRTV HQ and a regional station (e.g., CRTV Douala).

Result: Passed – Network traffic was rerouted through the backup link using HSRP/MPLS without service interruption.

➤ Connectivity Test Across Regions

Test Case: Test end-to-end connectivity between CRTV HQ and all five regional groups.

Result: Passed – All regions maintained communication with HQ via at least one active link.

➤ Firewall and Security Test

Test Case: Verify traffic filtering and access control from unauthorized networks.

Result: Passed – Firewall rules correctly blocked or allowed traffic as configured, ensuring data protection.

➤ Load and Redundancy Test

Test Case: Simulate heavy traffic between CRTV nodes and observe network stability.

Result: Passed – Routers and switches remained stable, with backup links managing the load during congestion.

➤ **GNS3 Simulation Integrity**

Test Case: Confirm that the GNS3 simulation reflects real-world failover and routing behavior.

Result: Passed – Simulation showed accurate router behavior, with minimal downtime during failure recovery.

CHAPTER 5: Discussion, Conclusion, and Recommendations

After exploring how CRTV implemented a backup network, this chapter looks at key results and provides an overall conclusion to the study. It examines how the backup system handles CRTV's problems linked to network downtime, failures of the main telecom system and supplying services. The section also delivers tips for boosting CRTV's network strength and making sure the network can continue to operate reliably into the future by using affordable and scalable backup resources.

5.1 Discussion of Findings

Building a stable backup network system for CRTV answered the urgent problems explained in the first chapter. The solution suggested represented by redundant links, automatic changeover methods and secured communication systems for data. The goal was to decrease downtime during services, shield data and guarantee regular broadcasts and communication within the company, even during serious network outages. The framework was built to be both cost-effective, simple to support and adaptable to CRTV's work and budget. Observations and interviews with network staff proved that the updated design improves the system's reliability, guarantees continuous operation and prepares for disruptions better.

5.1.1 Summary of Key Findings

Based on the data gathered from CRTV staff and the simulation of the proposed backup network, the following key findings were

1. Importance of Having a Strong Backup Network

Based on the study, it was found that CRTV's current network design is weak to interruptions and lacks backup protection. Observing and interviewing people proved that regular outages due to power failures, equipment breakdowns or cyber issues interrupted the company's style of internal communication as well as its communication with the public. T

A solution was created using several ways to reach the network and automatic changes if one path fails. As a result, if the main network goes down, business operations will face only a little

interruption. The infrastructure provides a high level of reliability by ensuring constant broadcasting for a nationwide broadcaster.

2. How the system works and how stable it is

During GNS3 simulations, the proposed backup network was highly reliable. For example, running dual routers, using backup switches and setting up automated routing (such as

The staff who reviewed the prototype judged that the system being developed provides a greater sense of stability and strength than what exists now. Since the design scales, CRTV can grow without having to change its entire system.

3. Challenges Identified

Nevertheless, a few difficulties were noticed when the design was developed.

Problems with Hardware and Power: At present, the software faces challenges due to CRTV's limited resources for hardware and backup power. Some devices might have to be upgraded to meet the requirements of the new backup design.

A lack of advanced networking knowledge among team members may make it harder for the company to use advanced systems. It underlines the need to keep receiving training and support after purchase.

4. Protecting Security and the Correctness of Your Data

A key problem that appeared was how to ensure security. Content files, transmission schedules and details on employees are handled by CRTV. Data on a network can be lost when there is no safe way to back it up.

As a strategy, the design consists of these measures:

Integration of firewalls and routed traffic encryption.

Use access control techniques to stop people who shouldn't be using the service.

Frequently backing up your data and safely setting up the main services.

This protection guarantees the security and confidentiality of CRTV's data during regular activities and during emergency changes.

5.2 Conclusion

The goal of this research was to build and test a backup network architecture that supports the operation of the main control system.

The research proposed building a secure and efficient backup network using Cisco's technologies and then simulation in GNS3 to look at the benefits. Important elements such as alternative links, automatic rerouting and protected routing protocols were included in the design for continuous network operation. The simulated tests showed that the backup system ensures network dependability and keeps CRTV's broadcasting and internal services running even if the main network fails.

Researchers found that it is still challenging to carry out CRTV tasks because of the current system's shortcomings and the desire for more improved technical abilities.

It was found that a resilient backup network can lower the downtime caused by CRTV, make operations smoother and ensure the uninterrupted provision of services. If IT policies, staff education and IT systems are correctly set up, CRTV can become more modern in its IT operations to support future demands of public broadcasting.

5.3 Recommendations

Based on the findings of this study, the following recommendations are proposed to improve the reliability, performance, and adoption of the backup network system at CRTV:

5.3.1 System Enhancement

- **Advanced Monitoring Tools**

The proposed backup network at CRTV should be upgraded with advanced monitoring systems that provide real-t

- **Technical Training for IT Staff**

One of the challenges observed was limited in-house expertise in handling network resilience syst

5.3.2 Future Research Directions

- **Scalability and Load Testing**

Future studies should assess how the proposed backup network performs under increasing user loads and higher traffic volumes. Testing across CRTV's regional offices or additional departments will help evaluate the system's

- **Integration with Cloud-Based Redundancy**

Research should also focus on integrating the local backup system with cloud-based disaster recovery services. This hybrid setup would allow CRTV to store critical data offsite and recover faster in case of physical infrastructure failure.

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